# Quantum Mechanics I

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# Syllabus

## Course structure

15 weeks, two 100-minute lectures per week. For each week, 6 problems (2+2+2 for three levels of difficulty), a list of suggested reading (including original and experimental sources), and a 1-page[1] summary sheet are provided. The midterm exam contributes 30%, the final exam 70%. The exam questions will be variants of the weekly problems.

## Course content (tentative)

| Wk | Theme | lectures: keywords |
|---|---|---|
| I | systems, states, measurement | $\aleph_I$: the need for a non-classical description of physical systems (listing by Dirac); the spin/polarization experiment as the generic example; summary of vector spaces and complex numbers; Dirac notation and how (dual) vectors, inner products, and outer products are represented; polarization of a photon as a "classical" example of a two-level, quantum system; |
| | | $\beth_I$: relation between states and vectors; spin states in Dirac notation, and how the observation of the spin/polarization measurement is described in this framework; postulates of quantum mechanics including the operator-measurement correspondence (matrix representation, eigenvalues); illustration with spin operators; |
| II | dynamics, symmetry | $\aleph_{II}$: discretized amplitude hopping; continuum limit; Schrödinger equation; wave function as amplitude for "acts as if"; position and momentum representations. |
| | | $\beth_{II}$: unitary transformation; Hermitian generator; translation, rotation, time; conservation as $[A, U] = 0$; stationary states; eigenvalue problem. |
| III | harmonic oscillator | $\aleph_{III}$: algebraic solution; $a, a^\dagger$; ladder; zero point; representation change as coordinate change. |
| | | $\beth_{III}$: coherent states; quadratures; phase space; the state that follows a classical trajectory, and why it is the exception. First promise redeemed in lecture XXVI. |

[1] Named after Paulo Bedaque who envisioned the perfect presentation in a 1-slide format to me.

| Wk | Theme | lectures: keywords |
|---|---|---|
| IV | angular momentum I | $\aleph_{IV}$: rotation generators; $\mathfrak{su}(2)$ algebra; ladder operators; spectrum from the algebra alone. |
| | | $\beth_{IV}$: matrix representations; spin-$\frac{1}{2}$ and the qubit identified; $2\pi$ rotation and the sign. |
| V | angular momentum II | $\aleph_{V}$: orbital angular momentum; spherical harmonics; parity. |
| | | $\beth_{V}$: rotation matrices; Wigner $d$; irreducible tensors; Wigner–Eckart. |
| VI | central forces | $\aleph_{VI}$: radial equation; centrifugal barrier; bound states; degeneracy. |
| | | $\beth_{VI}$: hydrogen; scaling; accidental degeneracy and its symmetry origin. |
| VII | approximation | $\aleph_{VII}$: time-independent perturbation theory, non-degenerate and degenerate; variational principle; Hellmann–Feynman. |
| | | $\beth_{VII}$: time-dependent perturbation theory; Fermi golden rule; adiabatic theorem; Berry phase. |
| VIII | magnetic fields | $\aleph_{VIII}$: minimal coupling; gauge; Landau levels. |
| | | $\beth_{VIII}$ : Zeeman (now with perturbation theory available); Aharonov–Bohm; geometric phase revisited. |
| IX | two-level dynamics | $\aleph_{IX}$: spin precession; Larmor; Rabi oscillation; rotating frame. |
| | | $\beth_{IX}$: pulses as gates; Bloch-sphere trajectories; NMR as the original quantum control. |
| X | coupling of angular momenta | $\aleph_{X}$: addition; Clebsch–Gordan; singlet and triplet. |
| | | $\beth_{X}$: fine and hyperfine structure; the 21 cm line; atomic clocks. |
| XI | mixed states | $\aleph_{XI}$: density matrix; purity; Bloch ball; reduced state; partial trace. |
| | | $\beth_{XI}$: von Neumann's measurement chain; where the cut is placed; decoherence and the preferred basis; Kraus maps and the Lindblad equation in outline. |
| XII | correlations | $\aleph_{XII}$: entanglement; Schmidt decomposition; monogamy; no-cloning. |
| | | $\beth_{XII}$: Bell inequality; CHSH; loopholes; what locality and realism each cost; teleportation. |
| XIII | **transformation theory** | $\aleph_{XIII}$: representation as a choice of basis; continuous spectra; transformation functions $\langle x\|p\rangle$; unitary equivalence; Schrödinger, Heisenberg and interaction pictures as one content in three dresses. |
| | | $\beth_{XIII}$: quantum conditions; Poisson bracket to commutator; canonical quantization and where it is ambiguous; Ehrenfest's theorem; coherent states revisited; in what sense a limit is classical. |
| XIV | **symmetry and conservation** | $\aleph_{XIV}$: symmetry groups; generators and Lie algebra; translation, rotation, Galilean boost, and the conserved quantity belonging to each; degeneracy as a group statement; Runge–Lenz and the hydrogen degeneracy of lecture XII explained. |
| | | $\beth_{XIV}$: Wigner's theorem, symmetries are unitary or antiunitary; parity; time reversal and its antiunitarity; Kramers degeneracy; permutation symmetry stated as a postulate and deliberately not developed. |
| XV | **action and path integral** | $\aleph_{XV}$: Dirac's 1933 paper; Feynman's sum over paths; the propagator; the free particle done completely; stationary phase and the emergence of the classical trajectory. |

| Wk | Theme | lectures: keywords |
|---|---|---|
| | | $\beth_{XV}$: the action principle; Noether and Schwinger; symmetry to conservation law inside the path integral; Aharonov–Bohm as a phase around a loop; what the advanced course opens with. |

# Scriptum

## $[\aleph_I]$ The essence of quantum behavior

A description in which objects carry their properties of which they can be asked at any time by pertinent measurements anytime is not how nature works. There are, however, objects which always *answer* certain questions with a definite answer, and we want to call those objects "large". In contrast, it was found very early that there is the other kind, the "small" system which does reply in a seemingly random way. For the latter, we want to devise a quantitative description. An all encompassing example is the behavior of light of extremely low intensity. The sought-after description must account for:

- Indivisibility,
- polarization,
- interference,
- superposition, and
- indeterminacy.

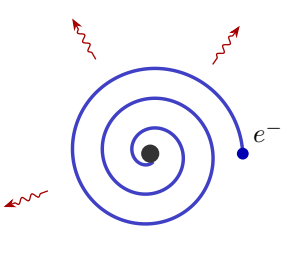
**And**, an explanation of Rutherford's scattering (1911) which goes beyond the "unstable" one suggesting the atom as a planetary system of electrons orbiting a heavy nucleus by means of electromagnetic attraction.

Larmor tells us that any accelerated charge radiates away power, and a classical lifetime estimate would predict the electrons to spiral onto the nucleus in a very short time.

Dimensional analysis gives an even farther-reaching indication: there must be something at work which the classical constants $m$, $e^2/4\pi\varepsilon_0$ and $c$ cannot quantify: They can be combined to yield

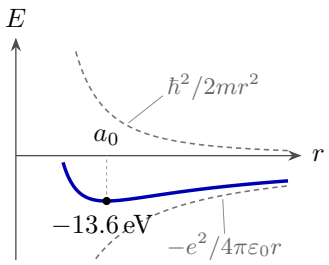
$$r_e = \frac{e^2}{4\pi\varepsilon_0 mc^2} = 2.8 \text{ fm},$$

while all other length scales are even smaller than this four-orders-of-magnitude-too-small one. In other words, there is no classical length the atomic size Rutherford was probing. [Estimate the order of magnitude and unit of a new constant of nature that could set the atomic scale $a_0 \approx 0.529$ Å.]



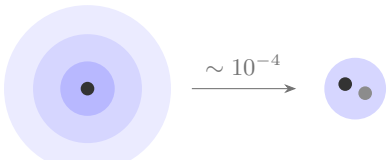

$t \simeq 1.6 \times 10^{-11}$ s

collapse due to radiation




a size, and a floor

stability demanding a new (quantum) scale




atom
0.5 Å, $-13.6$ eV

deuteron
2 fm, $-2.2$ MeV

quantum mechanism ensuring stability in atoms and nuclei

In turn, once a this novel mechanism introduces the new length scale, the stability of the atomic nucleus can/could be a consequence of the same mechanism.

But I want to start with a tangible example which should embody the key idea which represents the fundamental shift in philosophy that quantum mechanics introduces:

*Classical physics: a system has properties, and measuring reveals them.*

*Quantum physics: a system has a state; a measurement is a question; some questions are incompatible, and answering one erases the answer to another.*

It remains to specify (mathematically):

- How the state is represented? [vector]
- How is the state measured, in general, manipulated? [expectation values and operators]
- How is its evolution in time predicted? [Action principle → Schrödinger equation, &c.]

### measurements on two-state systems

Let us begin by putting on Stern and Gerlach's shoes, and send Atoms into an inhomogeneous magnetic field.

[SIM] `stern-gerlach.html`

More generally, this asks what is beeing detected if a beam of particles, presumably all in the same state, is sent through an appropriate version of a Stern-Gerlach apparatus? First, let us run the experiment such that we can trace each particle, silver atom or photon or ..., individually. We find

$$\sigma(\hat{n}) = \pm 1 \quad \text{for every filter orientation} \quad \hat{n} \ , \qquad \text{never } 0 \ , \qquad \text{never } \cos\theta \ .$$

[$\sigma(\hat{n})$: the reading of the apparatus whose axis points along the unit vector $\hat{n}$]

and a spatial splitting of the beam; the classically-expected band is not found.

What happens if the apparatus is rotated? Nothing, at all, for we would still detect only $\pm 1$ for each particle. Now, let us add another Stern-Gerlach apparatus in sequence, with its axis oriented along a different direction. The idea is to allow only one of the two states exiting the first filter to enter the second. This is what the above simulation does when the oven emits particles with a defined "orientation". Note that although we call it orientation, we do not yet know what we are dealing with except that "it" apparently dictates whether our atom ends up in the "up" or "down" path of the Stern-Gerlach apparatus.

You should now find what the two guys found long ago, namely not the

$$\underbrace{\sigma_{\text{cl}} = \hat{n} \cdot \hat{m} \in [-1, 1]}_{\text{expected, continuous}} \quad \text{but the discrete} \quad \sigma = \pm 1 \quad \text{with the special case} \quad \sigma\big|_{\hat{n}=\hat{m}} = +1 \quad .$$

[$\hat{m}$: the preparation (polarizer) axis, $\hat{n}$: the analysis (analyzer) axis, both unit 3-vectors in ordinary space.]

Discuss the most general arrangement: polarizer oriented in $\hat{m}$ and analyzer oriented in $\hat{n}$ direction. For a single particle, the outcome is $\pm 1$, for all (!) orientations of oven and filter (do not confuse the behind-the-filter blocking with the actual measurement outcome).

Considering an ensemble of identically prepared particles along $\hat{m}$, we observe an average outcome

$$\langle \sigma \rangle \ = \ \hat{n} \cdot \hat{m} \ = \ \cos\theta \ , \qquad \theta = \sphericalangle(\hat{m}, \hat{n}) \quad ,$$

[$\langle \cdot \rangle$: arithmetic mean of single-particle outcomes] which seems to measure the component of the preparation axis $\hat{m}$ along the analysis axis $\hat{n}$.

Let us now cast this in the language of probabilities: "How probable is it to get $\sigma = +1$ or $\sigma = -1$ for given '$\hat{m}$ and $\hat{n}$": [$P_\pm$: probability that the analyzer along $\hat{n}$ reads $\sigma = \pm 1$, for a particle prepared along $\hat{m}$. They sum to one because no third reading ever occurs.]

$$P_+ + P_- = 1\,, \qquad P_+ - P_- = \cos\theta \qquad \Longrightarrow \qquad P_\pm = \frac{1 \pm \cos\theta}{2}\,, \qquad P_+ = \cos^2\frac{\theta}{2}\,, \quad P_- = \sin^2\frac{\theta}{2}\,,$$

$$\theta = 0:\ (1,0)\ ; \qquad \theta = \tfrac{\pi}{2}:\ \left(\tfrac{1}{2},\tfrac{1}{2}\right)\ ; \qquad \theta = \pi:\ (0,1)\ .$$

Let us proceed with a setup designed to shed light on whether or not the measurements erase information encoded in the initial preparation along $\hat{z}$. With the sequence

$$\hat{z} \ \longrightarrow\ \hat{x} \ \longrightarrow\ \hat{z} \quad \text{we find} \quad \langle \sigma_z \rangle = 0 \quad ,$$

[$\sigma_z \equiv \sigma(\hat{z})$, and likewise $\sigma_x, \sigma_y$: the reading with the apparatus along a coordinate axis. The arrows are successive settings of one and the same apparatus, in time order.] and thus, if we allow only the $+\hat{z}$ channel to pass through $\hat{x}$, the right $\hat{z}$' result $\langle \sigma_z \rangle = 0$ implies equal probabilities for the $+\hat{z}$ and $-\hat{z}$ outcomes.

You can analyze this with

[SIM] `stern-gerlach-three.html`

which you (sh/c)ould parametrize such that it reproduces the original 2-filter experiment.

Having seen all this in a context (atoms) where a different outcome was expected, recall that the observed behavior is what is classically accepted when dealing with photons and their polarization. The latter is described by the three sets of orthogonal polarization states: vertical/horizontal, diagonal/anti-diagonal, and left-/right-circular, which correspond to the three ways the atoms leave the Stern-Gerlach apparatus along three mutually orthogonal axes (magnetic fields along $\hat{x}, \hat{y}, \hat{z}$). The following pairs represent the corresponding polarization directions of the electric/magnetic field of the photon:

$$\{\hat{v}, \hat{h}\}\ , \qquad \{\hat{d}, \hat{a}\}\ , \qquad \{\hat{r}, \hat{l}\}\ ,$$

[$\hat{v}, \hat{h}, \hat{d}, \hat{a}, \hat{r}, \hat{l}$: the six analyzer settings vertical, horizontal, diagonal, anti-diagonal, right-circular, left-circular.] and if a photon is polarized along any one of these, the probability to find it in any state of the other two pairs is $P = \frac{1}{2}$. If the angle between initial and analyzing polarizer is $\theta$, the probability to find the photon in the analyzing polarizer is given by Malus' law which Stern and Gerlach found for their atomic beams, too:

$$P_+ = \cos^2\theta \quad \text{and} \quad \langle \sigma \rangle = \cos 2\theta \quad .$$

These observations are now to be explained. They suffice to guess/postulate the formal structure of QM. First, let us get to the stage of knowledge from where we can make such a conjecture: what are complex numbers, and vector spaces (bra, kets, inner products, operators, and bases)?

### complex numbers

$$z = x + iy\ , \qquad x, y \in \mathbb{R}\ , \qquad i^2 = -1\ , \qquad\qquad z = r\,e^{i\theta} = r\,(\cos\theta + i\sin\theta)\ .$$

[$r = \sqrt{z^* z} \geq 0$ is called modulus, and $\theta$ argument. Two real numbers represent one complex number.] Addition is easier in components, multiplication in polar form:

$$z_1 + z_2 = (x_1 + x_2) + i\,(y_1 + y_2)\ , \qquad\qquad \left(r_1 e^{i\theta_1}\right)\left(r_2 e^{i\theta_2}\right) = r_1 r_2\, e^{i(\theta_1 + \theta_2)}\ .$$

Every $z$ has a unique conjugate/dual $z^*$, with $z^* z$ real and non-negative:

$$z = x + iy = r\,e^{i\theta} \quad \Longleftrightarrow \quad z^* = x - iy = r\,e^{-i\theta}\ , \qquad\qquad z^* z = r^2 \geq 0\ .$$

We denote those numbers with $r = 1$,

$$z^* z = 1 \ , \qquad z = e^{i\theta} = \cos\theta + i \sin\theta \quad ,$$

phase factors or simply phases.

**vector spaces**

The measurement outcomes discussed above suggest the following, profound switch:

$$\text{classical system} \leftrightarrow \text{set} \qquad \Leftrightarrow \qquad \text{quantum system} \leftrightarrow \text{vector}$$

Let us denote these state vectors "kets", and use the notation:

$$|\text{set of numbers which characterize the state}\rangle \stackrel{\text{e.g.}}{=} |\text{position, mass, (hair) color, (eye) color, \&c.}\rangle \quad .$$

If vectors, those objects must satisfy the axioms of a vector space:

(i) $|A\rangle + |B\rangle = |C\rangle$
(ii) $|A\rangle + |B\rangle = |B\rangle + |A\rangle$
(iii) $\big\{\, |A\rangle + |B\rangle \,\big\} + |C\rangle = |A\rangle + \big\{\, |B\rangle + |C\rangle \,\big\}$
(iv) $\exists!\ 0 \ : \quad |A\rangle + 0 = |A\rangle$
(v) $\exists!\ -|A\rangle \ : \quad |A\rangle + \big(-|A\rangle\big) = 0$
(vi) $|zA\rangle = z\,|A\rangle = |B\rangle$
(vii) $z\big\{\, |A\rangle + |B\rangle \,\big\} = z\,|A\rangle + z\,|B\rangle \ , \qquad \big\{z + w\big\}\,|A\rangle = z\,|A\rangle + w\,|A\rangle$

Two-dimensional state vectors can be represented concretely as columns of complex numbers:

$$|A\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \ , \qquad \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 \\ \alpha_2 + \beta_2 \end{pmatrix} \ , \qquad z \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} z\alpha_1 \\ z\alpha_2 \end{pmatrix} \ .$$

[$\alpha_i, \beta_i \in \mathbb{C}$: the components of $|A\rangle$ and $|B\rangle$ with respect to a basis that has been fixed beforehand while the abstract ket refers to the state itself without reference to any basis.]

To every "ket" belongs a "bra" with an *anti*linear correspondence:

$$|A\rangle \ \longleftrightarrow \ \langle A| \ , \qquad |A\rangle + |B\rangle \ \longleftrightarrow \ \langle A| + \langle B| \ , \qquad z\,|A\rangle \ \longleftrightarrow \ \langle A|\, z^* \ ,$$

$$|A\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix} \qquad \Longleftrightarrow \qquad \langle A| = \begin{pmatrix} \alpha_1^* & \cdots & \alpha_N^* \end{pmatrix} \ .$$

The inner product is always bra times ket, and returns a complex number:

$$\langle C| \big\{\, |A\rangle + |B\rangle \,\big\} = \langle C|A\rangle + \langle C|B\rangle \ , \qquad \langle B|A\rangle = \langle A|B\rangle^* \ ,$$

$$\langle B|A\rangle = \sum_i \beta_i^* \alpha_i \ , \qquad \langle A|A\rangle = 1 \ \ \text{(normalized)} \ , \qquad \langle B|A\rangle = 0 \ \ \text{(orthogonal)} \ .$$

The dimension of the space is the largest number of mutually orthogonal vectors in it; an orthonormal basis $|i\rangle$, $i = 1 \ldots N$, then resolves any ket, the components being its inner products with the basis vectors: [$\delta_{ij}$: the Kronecker symbol, 1 for $i = j$ and 0 otherwise — here it is precisely the statement that the basis is orthonormal.]

$$\langle j|i\rangle = \delta_{ij} \ , \qquad |A\rangle = \sum_i \alpha_i\,|i\rangle \ , \qquad \alpha_j = \langle j|A\rangle \ , \qquad |A\rangle = \sum_i |i\rangle\,\langle i|A\rangle \ .$$

## $[\beth_I]$ Mapping physics onto mathematics

Now let us map physics onto math: the state ket-vector identification (Dirac and row/column notation). Specifically, spin states: all there is to know, there is to measure, are the components along a given direction in space, and since ay such direction in 3 dimensions is specified by two numbers, the state space is two-dimensional. Express the measurement outcomes for the xyz directions using Dirac notation $|\uparrow_x\rangle, |\downarrow_x\rangle, |\uparrow_y\rangle, |\downarrow_y\rangle, |\uparrow_z\rangle, |\downarrow_z\rangle$ and phrase that in terms of three pairs of mutually orthogonal vectors whose projections on one another are identical. As a consequence: the vector space is complex and two-dimensional.

## $[R_I]$ Reading list


[1] O. Stern, "A Way towards the Experimental Examination of Spatial Quantisation in a Magnetic Field", trans. by E. E. B. Campbell and B. Friedrich, Zeitschrift für Physik D: Atoms, Molecules and Clusters **10**, 114–116 (1988) doi:10.1007/BF01384842; trans. of "Ein Weg zur experimentellen Prüfung der Richtungsquantelung im Magnetfeld", Zeitschrift für Physik **7**, 249–253 (1921) doi:10.1007/BF01332793.

[2] W. Gerlach and O. Stern, *The Stern-Gerlach Experiment*, trans. by M. Bauer, (2023) arXiv:2301.11343 [physics.hist-ph]; trans. of "Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld", Zeitschrift für Physik **9**, 349–352 (1922) doi:10.1007/BF01326983.

[3] G. E. Uhlenbeck and S. Goudsmit, "Spinning Electrons and the Structure of Spectra", Nature **117**, 264–265 (1926) doi:10.1038/117264a0.

[4] R. P. Feynman, F. L. Vernon, and R. W. Hellwarth, "Geometrical Representation of the Schrödinger Equation for Solving Maser Problems", Journal of Applied Physics **28**, 49–52 (1957) doi:10.1063/1.1722572.

[5] L. Susskind and A. Friedman, *Quantum Mechanics: The Theoretical Minimum* (Basic Books, 2014), chapters 1–3 [pedagogical].

[6] G. Baym, *Lectures on Quantum Mechanics* (W. A. Benjamin, 1969), chapter 1 up to "Angular Momentum" [pedagogical].


## $[P_I]$ Problem list

### Level 1

**P 1. Three Stern-Gerlach filters ($\mathrm{SG}_i$).** A beam leaves the oven with no preparation axis. It passes $\mathrm{SG}_1$ along $\hat{z}$, of which only the $\sigma = +1$ channel continues; then $\mathrm{SG}_2$ along $\hat{n}$ at an angle $\theta$ to $\hat{z}$, again keeping only $\sigma = +1$; then $\mathrm{SG}_3$ along $\hat{z}$.

(a) Applying $P_\pm(\theta) = \cos^2\frac{\theta}{2},\ \sin^2\frac{\theta}{2}$ once at each analyser, give the fraction of the oven output reaching each of the two $\mathrm{SG}_3$ channels.

(b) Evaluate at $\theta = 0,\ \pi/2,\ \pi$. At which $\theta$ is the $\sigma_z = -1$ count largest, and what is it?

(c) Remove $\mathrm{SG}_2$ and give the same two numbers.

(d) Suppose every atom carried a fixed label $\lambda$ and each analyser passed or stopped it according to $\lambda$ alone. Transmission would then be intersection of sets. Show $T_{123} = T_1 \cap T_2 \cap T_3 \subseteq T_1 \cap T_3$, so inserting $\mathrm{SG}_2$ cannot raise any count. Compare with (b) and (c).

[SIM] `stern-gerlach-three.html` runs the chain; check your (a) against the counter before you trust either.

**P 2. Measurements on single photons and on ensembles.** A polarizer and an analyser have their transmission axes at $\theta_{\text{lab}}$.

(a) For a beam, $I/I_0 = \cos^2\theta_{\text{lab}}$; for one photon, $P = \cos^2\theta_{\text{lab}}$. The same function is being used twice. State precisely what each of the two equations asserts, and what is being counted in each.

(b) $N$ photons are sent one at a time. The transmitted number $K$ is binomial with $p = \cos^2\theta_{\text{lab}}$. Give $\langle K\rangle$ and $\operatorname{Var} K$.

(c) Show $\sqrt{\operatorname{Var} K}/\langle K\rangle = \sqrt{(1-p)/(Np)}$. At $\theta_{\text{lab}} = 45°$, how large must $N$ be before the measured transmission matches Malus to one percent?

(d) Attenuating a laser until on average one photon per second arrives does *not* make it a one-photon source. Say why, and name a measurement that separates the two.

## Level 2

**P 3. The three bases.** In $\mathbb{C}^2$ fix an orthonormal pair $|z+\rangle, |z-\rangle$ and define

$$|x\pm\rangle = \tfrac{1}{\sqrt{2}}\left(|z+\rangle \pm |z-\rangle\right), \qquad |y\pm\rangle = \tfrac{1}{\sqrt{2}}\left(|z+\rangle \pm i\,|z-\rangle\right).$$

(a) Verify that each of the three pairs is orthonormal.

(b) Compute $\left|\langle a|b\rangle\right|^2$ for every $a$ taken from one pair and $b$ from another. Show that all twelve numbers equal $\frac{1}{2}$.

(c) The photon bases $\{|H\rangle, |V\rangle\}$, $\{|D\rangle, |A\rangle\}$, $\{|R\rangle, |L\rangle\}$ obey the same relations. Write $|D\rangle, |A\rangle, |R\rangle, |L\rangle$ in terms of $|H\rangle, |V\rangle$ and repeat (b) in that language.

(d) On the sphere a basis is an axis, and $\left|\langle a|b\rangle\right|^2 = \cos^2\frac{\theta}{2}$ with $\theta$ the angle between axes. Show $\left|\langle a|b\rangle\right|^2 = \frac{1}{2}$ holds exactly when the axes are perpendicular, and conclude that $\mathbb{C}^2$ admits at most three such bases.

**P 4. discreteness from an anti-commutator.** Let $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ be the Pauli matrices and $\hat{n}$ a real unit vector with polar angles $(\theta, \phi)$.

(a) Write $\sigma_n = \hat{n}\cdot\vec{\sigma}$ as a $2\times 2$ matrix.

(b) Using $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}\mathbf{1}$, show $\sigma_n^2 = \mathbf{1}$ for every $\hat{n}$. Deduce that the only possible readings are $\pm 1$ — without solving an eigenvalue problem.

(c) Now solve for the eigenvectors and show, up to an overall phase,

$$|\hat{n}+\rangle = \cos\tfrac{\theta}{2}\,|z+\rangle + e^{i\phi}\sin\tfrac{\theta}{2}\,|z-\rangle, \qquad |\hat{n}-\rangle = -\sin\tfrac{\theta}{2}\,|z+\rangle + e^{i\phi}\cos\tfrac{\theta}{2}\,|z-\rangle.$$

(d) Compute $\left|\langle \hat{n}+|z+\rangle\right|^2$ and recover $\cos^2\frac{\theta}{2}$.

(e) In the state $|\hat{m}+\rangle$, compute $\langle\sigma_n\rangle$ and show it equals $\hat{n}\cdot\hat{m}$.

[Susskind Lecture 3, Exercises 3.3 and 3.4, worked in his §3.7.]

## Level 3

**P 5. The Bloch sphere.**

(a) Show that $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ give identical probabilities for every measurement, so the physical state is the ray and not the vector.

(b) A normalised vector in $\mathbb{C}^2$ carries four real numbers; normalisation and the global phase remove two. Show that $|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle$ with $\theta \in [0,\pi]$, $\phi \in [0,2\pi)$ covers every ray exactly once. Mind the poles.

(c) Show that $\langle\vec{\sigma}\rangle = \hat{n}$ in the state $|\hat{n}+\rangle$: the Bloch vector is the expectation value of $\vec{\sigma}$.

(d) Show that orthogonal states are antipodal, and mark the six states of P 3 on a sphere.

(e) A polarization ellipse returns to itself after a laboratory rotation of $\pi$, while the state vector returns only after $2\pi$ and collects a minus sign at $\pi$. Explain how both are true at once, and connect it to the $2\pi$ rotation of a spin-$\frac{1}{2}$.

(f) Why is the set of rays in $\mathbb{C}^2$ called $\mathbb{CP}^1$, and why is $\mathbb{CP}^1 \cong S^2$? Build the stereographic map explicitly.

**P 6. Labels as an alternative model.** A concrete classical model: each atom leaves the oven carrying a unit vector $\vec{\lambda}$, isotropically distributed; an analyser along $\hat{n}$ reports $\sigma = \mathrm{sign}(\vec{\lambda}\cdot\hat{n})$; a filter keeps the atoms it reports $+1$ for and discards the rest, doing nothing else to $\vec{\lambda}$.

(a) For the unprepared oven beam, show $P_\pm = \frac{1}{2}$ for every $\hat{n}$ and $\langle\sigma\rangle = 0$. The model survives this test — which is why the first experiment alone decides nothing.

(b) Prepare by keeping the + channel along $\hat{m}$, so $\vec{\lambda}\cdot\hat{m} > 0$, and then measure along $\hat{n}$ at angle $\theta$. Show that the two hemispheres overlap in a lune and hence

$$P_+ = 1 - \frac{\theta}{\pi}\,, \qquad \langle\sigma\rangle_{\text{model}} = 1 - \frac{2\theta}{\pi}\,.$$

(c) Compare with the measured $\langle\sigma\rangle = \cos\theta$. At which $\theta$ do the two agree exactly? Evaluate both at $\theta = \pi/3$, and find the $\theta$ at which they differ most.

(d) Apply the model to the chain of P 1 and show that it gives *zero* in the $\sigma_z = -1$ channel for every $\theta$. Identify the single assumption responsible, and state what the quantum account does in its place.

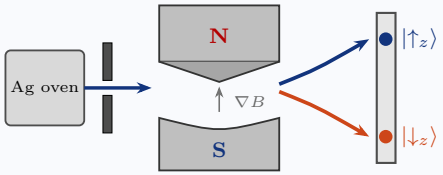
## $[T_I]$ The I-never-read-textbooks issues

- Isn't it surprising that we have already the means to predict the expected/average outcome of a system prepared in a certain state without having specified what the measurement does to the system? There is a measurement operator, yes, but that tells us only what kind of measurements are possible, and how likely those are (eigensystem). However, what state the system is left in post-measurement has not been discussed. But you should have given enough information to make a conjecture which state the system will be in if a particular measurement outcome is obtained.

$[C_I]$ **Paulo slide**

# Quantum Mechanics I – Week I: 2-state systems and the Bloch Sphere

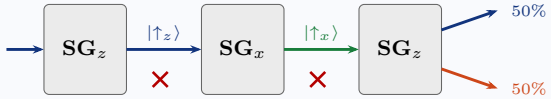
## 2- state atoms: Spin-($1/2$)




**Measurement outcomes**

- classical: $\vec{\mu}\cdot\hat{n}\in[-1,1]$, continuous
- **quantum: $\sigma(\hat{n})=\pm1$, discrete and $\hat{n}$ independent**

### Sequential filtering: incompatibility




**State erasure: $\hat{x}$ wipes out the prior $\hat{z}$**

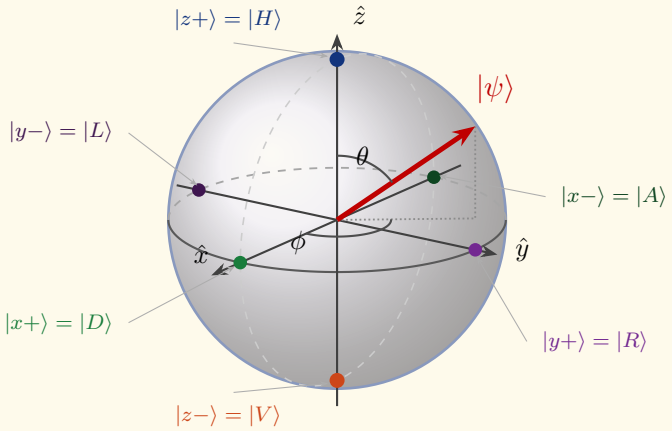
Without $\mathrm{SG}_x$ no counts in the lower channel. Adding $\mathrm{SG}_x$ counts as many clicks in the lower channel as in the upper channel.

$$T^{\downarrow/\uparrow}_{zxz}=\tfrac{1}{4}\neq0\Rightarrow\quad\langle\sigma_z\rangle=0\quad\textbf{vs.}\quad T^{\downarrow}_{zz}=0$$

$$p_\pm(\theta)=\cos^2\tfrac{\theta}{2},\ \sin^2\tfrac{\theta}{2}\qquad\langle\sigma\rangle=\hat{n}\cdot\hat{m}=\cos\theta$$

*$\theta=\sphericalangle(\hat{m},\hat{n})$: angle between preparing and analyzing axes.*

## 2-state systems: Bloch sphere




$$|\psi\rangle=\cos\tfrac{\theta}{2}\,|0\rangle+e^{i\phi}\sin\tfrac{\theta}{2}\,|1\rangle$$

$\{|z\pm\rangle\}=\{|H\rangle,|V\rangle\}$ $\quad\{|x\pm\rangle\}=\{|D\rangle,|A\rangle\}$ $\quad\{|y\pm\rangle\}=\{|R\rangle,|L\rangle\}$

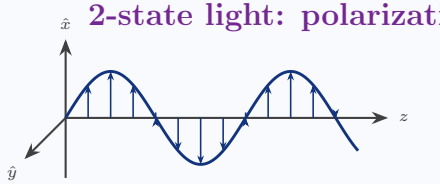
| the two states are | angle apart | $\lvert\langle a\vert b\rangle\rvert^2$ | SG reads |
|---|---|---|---|
| the same state | 0° | 1 | always + |
| partners in one basis | 180° | 0 | never |
| from two bases | 90° | 1/2 | 50/50 |

$$\left|\langle\hat{n}\,s|\hat{m}\,s'\rangle\right|^2=\tfrac{1}{2}\left(1+s\,s'\,\hat{n}\cdot\hat{m}\right)$$

$s,s'=\pm1,\quad\hat{n}\cdot\hat{m}=\cos\theta_{\text{Bloch}}=\cos2\theta_{\text{lab}}\ \Rightarrow\ \cos^2\tfrac{\theta}{2},\ \sin^2\tfrac{\theta}{2}$

*Orthogonal states are antipodal; a basis is an axis; axes are perpendicular.*

## 2-state light: polarization

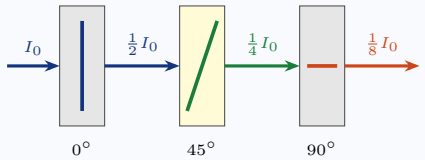



$$\vec{E}(z,t)=\mathrm{Re}\left[\vec{E}_0e^{i(kz-\omega t)}\right]$$

$$\vec{E}_0=E_0\left(\cos\alpha\,\hat{x}+e^{i\delta}\sin\alpha\,\hat{y}\right)$$

| basis | $\alpha$ | $\delta$ | $\vec{E}_0/E_0$ |
|---|---|---|---|
| $\lvert H\rangle,\lvert V\rangle$ | $0,\ \frac{\pi}{2}$ | – | $\hat{x}\,;\ \hat{y}$ |
| $\lvert D\rangle,\lvert A\rangle$ | $\frac{\pi}{4}$ | $0,\ \pi$ | $\frac{1}{\sqrt{2}}(\hat{x}\pm\hat{y})$ |
| $\lvert R\rangle,\lvert L\rangle$ | $\frac{\pi}{4}$ | $\pm\frac{\pi}{2}$ | $\frac{1}{\sqrt{2}}(\hat{x}\pm i\,\hat{y})$ |

### The three-filter experiment




**Single photon clicks**

- the detector registers indivisible clicks $\hbar\omega$
- Malus' law governs the individual probability:

$$p(\textbf{click})=\left|\langle\textbf{filter}|\psi\rangle\right|^2=\cos^2\theta_{\textbf{lab}}$$

- additional filter raises a count:

$$T_{0^\circ\to45^\circ\to90^\circ}=\tfrac{1}{8}\neq0\quad\textbf{vs.}\quad T_{0^\circ\to90^\circ}=0$$

## 1. HILBERT SPACE $\mathbb{C}^2$

$$|\psi\rangle=c_0\,|0\rangle+c_1\,|1\rangle,\quad|c_0|^2+|c_1|^2=1$$

Dual bra: $\langle\psi|=c_0^*\,\langle0|+c_1^*\,\langle1|$

Inner product: $\langle\phi|\psi\rangle=\phi_0^*\psi_0+\phi_1^*\psi_1\overset{!}{=}\langle\psi|\phi\rangle^*$

## 2. OBSERVABLES AS OPERATORS

$$A=A^\dagger=\textstyle\sum_i a_i\,|i\rangle\,\langle i|,\quad a_i\in\mathbb{R}$$

Eigenvalues = outcomes ($a_i=\pm1$)

Spin: $A=\vec{\sigma}\cdot\hat{n}$

$$\sigma_x=\begin{pmatrix}0&1\\1&0\end{pmatrix},\ \sigma_y=\begin{pmatrix}0&-i\\i&0\end{pmatrix},\ \sigma_z=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$$

## 3. BORN RULE & PROJECTORS

$$p_i=\left|\langle i|\psi\rangle\right|^2=\langle\psi|i\rangle\,\langle i|\psi\rangle=\langle\psi|P_i|\psi\rangle$$

Projectors: $P_i=|i\rangle\,\langle i|,\quad P_i^2=P_i=P_i^\dagger$

Identity: $\sum_i P_i=\mathbf{1}$

Expectation: $\langle A\rangle=\langle\psi|A|\psi\rangle=\sum_i a_ip_i$

## 4. MEASUREMENT COLLAPSE

$$|\psi\rangle\xrightarrow{\text{outcome }i}\frac{P_i\,|\psi\rangle}{\|P_i\,|\psi\rangle\,\|}=|i\rangle$$

Post-measurement state is eigenstate $|i\rangle$

Measurement prepares the state

# $[\aleph_{II}]$ Time development of a state vector

After having defined the mathematical representation of a quantum system whose state can assume any element of a state space which is a complex vector space, and after having defined how measurements reveal properties of the system in a specific state (nothing has been said yet about the state of the system post-measurement!), a complete theory must also describe the temporal evolution of the system.

I will present this evolution in the abstract on a general state vector $|\Psi(t)\rangle$, and for the specific case when this state vector does describe the spatial location of the system. Thus far, we only dealt with a 2-state system, and you need to generalize this notion such that a measurement can not only return two outcomes corresponding to up/down, for instance, but any location along a line or in space, in general.

It is a reasonable assumption that the evolution of a state from $t = 0$ to $t > 0$ is represented by a linear operator:

$$|\Psi(t)\rangle = U(t)\,|\Psi(0)\rangle \quad . \tag{4.1}$$

The first constraint we want to impose on $U(t)$ makes it preserve the distinctness of states:

$$\begin{aligned} \langle\Psi(0)|\Phi(0)\rangle = 0 \quad , \\ \langle\Psi(t)|\Phi(t)\rangle = 0 \quad . \end{aligned} \tag{4.2}$$

From this requirement, we can infer:

$$\langle\Psi(t)| = \langle\Psi(0)|\,U^\dagger(t) \tag{4.3}$$

$$\langle\Psi(0)|U^\dagger(t)U(t)|\Phi(0)\rangle = 0 \quad . \tag{4.4}$$

If we take the states to be orthonormal, e.g., as elements of an orthonormal basis, we get:

$$\begin{gathered} \langle i|j\rangle = \delta_{ij} \\ \langle i|U^\dagger(t)U(t)|j\rangle = 0 \qquad (i \neq j) \\ \langle i|U^\dagger(t)U(t)|j\rangle = \delta_{ij} \end{gathered}$$

and it follows that $U(t)$ must be a so-called unitary operator:

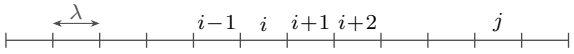
$$U^\dagger U = I$$

Let us now consider infinitesimally small time increments for which

$$U^\dagger(\epsilon)U(\epsilon) = I \tag{4.5}$$

holds, too, but we can also expand

Instead of $|\Psi(t)\rangle$ being a linear combination of vertical/horizontal, or diagonal/anti-diagonal, or right/left states at each time, let us imagine a system which can occupy any bin of width $\lambda$ along a one-dimensional line.




As we are describing a particle, it must be somewhere which implies that the sum over the squared overlap of the particle's state with the state $|i\rangle$ corresponding to bin $i$, $\psi_i(t) \equiv \langle i|\Psi(t)\rangle$, must be unity:

$$\sum_i |\psi_i(t)|^2 = 1 \tag{3-1}$$

Instead of the 2-component column vector, we are now dealing with the $\infty$-component column

$$|\Psi\rangle = \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \\ \vdots \\ \psi_{i-1}(t) \\ \psi_i(t) \\ \psi_{i+1}(t) \\ \vdots \end{pmatrix} \tag{3-2}$$

It is intuitive to think of time evolution of the state through leakage of overlap (probability amplitude) from bin to bin, and to assume (why?) that the amount of this leakage from bin $i$ to bin $i+1$ in $\Delta t$ is

$$-i\,\Delta t\,w_{i+1,i}\,\psi_i(t) \tag{3-3}$$

with $(-i)$ being factored out of the coefficients $w_{i+1,i}$, $w_{i,i-1}$, $w_{i,i+1}$, and $w_{ii}$ to make them furnish a hermitian matrix. It is now straightforward to represent the amplitude after a time increment $\Delta t$ by

$$U(\epsilon) = I - i\epsilon H \tag{4.6}$$
$$U^\dagger(\epsilon) = I + i\epsilon H^\dagger \tag{4.7}$$

and find that unitarity of $U(\epsilon)$ requires $H$ to be Hermitian:

$$\begin{aligned}(I + i\epsilon H^\dagger)(I - i\epsilon H) &= I \\ H^\dagger - H &= 0 \\ H^\dagger &= H \end{aligned} \tag{4.8}$$

The infinitesimal evolution of the state vector is expressed in terms of $H$:

$$\begin{aligned}|\Psi(\epsilon)\rangle &= |\Psi(0)\rangle - i\epsilon H\,|\Psi(0)\rangle \\ \frac{|\Psi(\epsilon)\rangle - |\Psi(0)\rangle}{\epsilon} &= -iH\,|\Psi(0)\rangle\end{aligned}$$

and in the limit $\epsilon \to 0$, we obtain the time-dependent Schrödinger equation:

$$\frac{\partial\,|\Psi\rangle}{\partial t} = -iH\,|\Psi\rangle \tag{4.9}$$

This equation implies that the operator $H$ has the dimensions of inverse time, but in order to have units of energy in conventional units, a constant with dimensions $\texttt{mass} \cdot \texttt{length}^2/\texttt{time}$ must be introduced. This constant is Planck's constant $h$, whose numerical value is chosen to yield numbers of order unity for typical quantum mechanical systems:

$$\begin{aligned}h &\approx 6.6 \times 10^{-34}\ \mathrm{kg\,m^2/s} \\ \hbar = \frac{h}{2\pi} &= 1.054571726\cdots \times 10^{-34}\ \mathrm{kg\,m^2/s} \\ \left[\frac{\partial}{\partial t}\right] = \mathrm{s}^{-1}, &\qquad [H] = \mathrm{kg\,m^2/s^2}, \qquad [\hbar] = \mathrm{kg\,m^2/s}\end{aligned}$$

$$\hbar\,\frac{\partial\,|\Psi\rangle}{\partial t} = -iH\,|\Psi\rangle \tag{4.10}$$

$$\begin{aligned}\psi_i(t+\Delta t) = \psi_i(t) &- i\,\Delta t\,w_{i,i-1}\psi_{i-1}(t) \\ &- i\,\Delta t\,w_{i,i+1}\psi_{i+1}(t) \\ &- i\,\Delta t\,\underbrace{(-w_{i,i+1} - w_{i,i-1} + v_i)}_{=:w_{ii}}\psi_i(t)\end{aligned} \tag{3-4}$$

with the first term representing what has been there at $t$, the second and third terms influx from left and right. The term $\propto \psi_i(t)$ represents outflux from bin $i$ and repulsion/attraction due to the "potential" of site $i$, $v_i$.

First, let us examine constraints on the bin-to-bin transition rates as they follow from

$$\sum_i |\psi_i(t+\Delta t)|^2 = \sum_i |\psi_i(t)|^2 \tag{3-5}$$

$$\begin{aligned}&\sum_i |\psi_i(t)|^2\big(w_{ii} - w_{ii}^*\big) \\ &+ \sum_i \psi_i^*(t)\psi_{i+1}(t)\big(w_{i,i+1} - w_{i+1,i}^*\big) \\ &+ \sum_i \psi_i^*(t)\psi_{i-1}(t)\big(w_{i,i-1} - w_{i-1,i}^*\big) = 0\end{aligned} \tag{3-6}$$

which yields:

$$w_{ii} = w_{ii}^*\,, \qquad w_{i,i+1} = w_{i+1,i}^* \tag{3-7}$$

In the more practical matrix notation, we can write this as

$$|\psi(t+\Delta t)\rangle = \big(1 - i\,\Delta t\,W\big)\,|\psi(t)\rangle \tag{3-9}$$

with

$$W = \begin{pmatrix} w_{11} & w_{12} & & & \\ w_{12}^* & w_{22} & w_{23} & & \\ & w_{23}^* & w_{33} & w_{34} & \\ & & w_{34}^* & w_{44} & w_{45} \\ & & & \ddots & \ddots \end{pmatrix} \tag{3-8}$$

Let us now take the continuous-time limit in Eq. (3-4):

$$i\,\frac{\partial\psi_i(t)}{\partial t} = w_{i,i-1}\psi_{i-1}(t) + w_{i,i+1}\psi_{i+1}(t) + w_{ii}\psi_i(t)$$

and resolve $w_{ii}$ such that one gets a very conspicuous form

$$\begin{aligned}i\,\frac{\partial\psi_i(t)}{\partial t} = w_{i,i-1}&\big[\psi_{i-1}(t) - \psi_i(t)\big] \\ &+ w_{i,i+1}\big[\psi_{i+1}(t) - \psi_i(t)\big] + v_i\,\psi_i(t)\end{aligned} \tag{3-12}$$

which reads in matrix notation

$$i\,\frac{\partial}{\partial t}\,|\Psi(t)\rangle = W\,|\Psi(t)\rangle$$

So in both columns, we arrive at the same structure for a differential equation. On the right, we can make the analogy with the heat equation, or any diffusion-like process characterized by the diffusion coefficient $\kappa$:

$$\frac{\partial T(\mathbf{r},t)}{\partial t} = \kappa\,\nabla^2 T(\mathbf{r},t)$$

Let us elaborate a little on this analogy as I find it easier to think of actual stuff flowing instead of "probability amplitudes". So we are allowed to identify $\psi_i(t)$ with some actual quantity, e.g., air, that flows. In a stationary situation without any inflow or outflow, we have can solve for $\psi_i(t)$ as

$$\psi_i(t) = e^{-iv_i t/\hbar}\,\psi_i(0) \tag{3-13}$$

As air flows continuously, we would like to take the limit $\lambda \to 0$ during which we need to be careful with the scaling of the discrete (unit-less because its square yields a dim-less probability) vs. continuous (with units of length$^{-1/2}$ as its square integrated over the line now yields the probability) representations:

$$\begin{aligned} \psi_i \;&\to\; \sqrt{\lambda}\,\psi(x_i) \\ [\psi_i] = 1\,, \qquad &[\psi(x)] = (\text{length})^{-1/2} \end{aligned}$$

Analogously to taking $\Delta t \to 0$, we replace the indices with arguments, and take the continuous limit $\lambda \to 0$:

$$i\,\frac{\partial}{\partial t}\psi(x_i,t) = w_{i,i-1}\big[\psi(x_{i-1},t) - \psi(x_i,t)\big] + w_{i,i+1}\big[\psi(x_{i+1},t) - \psi(x_i,t)\big] + v(x_i,t)\,\psi(x_i,t)$$

with $x_{i+1} = x_i + \lambda$ and the ensuing expansion

$$\psi(x_{i\pm1},t) = \psi(x_i,t) \pm \lambda\,\frac{\partial\psi(x_i,t)}{\partial x_i} + \frac{\lambda^2}{2}\,\frac{\partial^2\psi(x_i,t)}{\partial x_i^2} + \cdots \tag{3-15}$$

we obtain

$$i\,\frac{\partial}{\partial t}\psi(x_i,t) = \lambda\big[w_{i,i+1} - w_{i,i-1}\big]\frac{\partial\psi(x_i,t)}{\partial x_i} + \frac{\lambda^2}{2}\big[w_{i,i+1} + w_{i,i-1}\big]\frac{\partial^2\psi(x_i,t)}{\partial x_i^2} + \frac{v(x_i,t)}{\hbar}\,\psi(x_i,t)$$

which we can rewrite further by introducing strength functions characterizing the likelihood to hop to the left or to the right into any position $x$ along the line.

$$w_{i,i+1} = w_L(x_i)\,, \qquad w_{i,i-1} = w_R(x_i) \tag{3-17}$$

$$w_R(x) = w_L(x)^* \tag{3-18}$$

Note that this does not mean that the hopping rates to the left and right are equal. The complex conjugation resembles an inversion of direction (see Eq. Eq. (3-7)).

$$i\,\frac{\partial\psi(x,t)}{\partial t} = \frac{\lambda^2}{2}\big[w_L(x)+w_R(x)\big]\frac{\partial^2\psi(x,t)}{\partial x^2} + \lambda\big[w_L(x)-w_R(x)\big]\frac{\partial\psi(x,t)}{\partial x} + \frac{1}{\hbar}\,v(x,t)\psi(x,t) \quad \text{(3-19)}$$

However, unequal hopping corresponds to a preference for hopping in one direction over the other due to the property of the line. If we believe that this is the frictional force often disregarded in first approximation, we would set

$$w_R(x) = w_L(x) = w \quad .$$

Let us now be concrete, and try to relate this hopping rate to characteristics of the system we are describing, i.e., a single point particle in one spatial dimension subjected to nothing but the ominous tendency to hop, and which were are more familiar with.

To this end, let us find a solution to Eq. (3-19) in the case of a free particle, i.e., when $v(x,t) = 0$, and

$$i\,\frac{\partial\psi(x,t)}{\partial t} = \frac{\lambda^2}{2}w\frac{\partial^2\psi(x,t)}{\partial x^2}$$

For this equation, the freely oscillating plane wave is given by

$$\psi(x,t) = e^{i(kx-Et)}$$

and represents a solution if

$$E = \frac{\lambda^2}{2}wk^2$$

Now for the photon we know that $p = \hbar k$ and that in some limit which takes us from quantum to classical, we should obtain $E = \frac{p^2}{2m}$. Hence we identify $\frac{\lambda^2}{2}w = \frac{\hbar^2}{2m}$ and obtain the equation of motion for a quantum particle with mass $m$ in one spatial dimension:

$$i\hbar\,\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + v(x,t)\psi(x,t) \quad .$$

### Expectation values and their time evolution

We learned about the probability that a single measurement yields a particular eigenvalue. Now if we perform multiple measurements on identically prepared systems, we *expect* an outcome:

$$\langle L\rangle = \sum_i \lambda_i\,P(\lambda_i) \quad .$$

The state in which the system is prepared can be expanded in terms of the eigenstates of $L$:

$$|A\rangle = \sum_i \alpha_i\,|\lambda_i\rangle \quad ,$$

and we find nothing new by applying the operator $L$ to the state $|A\rangle$:

$$L\,|A\rangle = \sum_i \alpha_i\,L\,|\lambda_i\rangle = \sum_i \alpha_i\lambda_i\,|\lambda_i\rangle \quad .$$

In the above notation for the expectation value of $L$, the state in which this value is calculated is not explicit, but we must remember that it depends on that particular state:

$$\langle L\rangle = \langle A|L|A\rangle = \sum_i \left(\alpha_i^* \alpha_i\right)\lambda_i \quad .$$

Let us now consider how this quantity evolves with time:

$$\frac{d}{dt}\langle\Psi(t)|L|\Psi(t)\rangle = \langle\dot{\Psi}(t)|L|\Psi(t)\rangle + \langle\Psi(t)|L|\dot{\Psi}(t)\rangle \quad ;$$

Using the time-dependent Schrödinger equation, we obtain

$$\frac{d}{dt}\langle\Psi(t)|L|\Psi(t)\rangle = \frac{i}{\hbar}\langle\Psi(t)|HL|\Psi(t)\rangle - \frac{i}{\hbar}\langle\Psi(t)|LH|\Psi(t)\rangle \quad ,$$

which can be rewritten by introducing the so-called commutator of $H$ and $L$:

$$\frac{d}{dt}\langle\Psi(t)|L|\Psi(t)\rangle = \frac{i}{\hbar}\langle\Psi(t)|(HL - LH)|\Psi(t)\rangle \quad \text{, with}$$

$$LM - ML = [L, M] \quad . \qquad \text{(Commutator)}$$

In order to interpret

$$\frac{d}{dt}\langle L\rangle = \frac{i}{\hbar}\langle[H, L]\rangle \quad ,$$

we remember that $H$ is also a Hermitian operator and represents an observable, specifically the total energy of the system, as we have seen above in case of the free particle. Now we found that if the order of acting with $H$ and $L$ does not matter, i.e., if $[H, L] = 0$, the expected value of $L$ remains constant in time.

### Spin precision

Let us now apply this general formalism for a particle with spin-$\frac{1}{2}$, charge $e$, and mass $m_e$ as it evolves in the presence of a magnetic field. For this specific problem, it was found–although you can also derive this form from more general principles, it is mostly by observation and guesswork that one finds the appropriate structure for the Hamiltonian–that the Hamiltonian takes the form

$$H = -\left(\frac{e}{m_e c}\right)\boldsymbol{\sigma}\cdot\mathbf{B} \quad ,$$

and for the specific case where the magnetic field is along the $z$-axis:

$$H = -\left(\frac{eB}{m_e c}\right)\sigma_z \equiv \omega\sigma_z \quad .$$

Eigenstates are thus the familiar spin-up and spin-down states along the $z$-axis, and the eigenvalues are just multiplied by $-\frac{eB}{m_e c}$:

$$E_\pm = \mp\frac{e\hbar B}{2m_e c}\,, \qquad \text{for } H\,|z\pm\rangle = E_\pm\,|z\pm\rangle \quad .$$

In order to determine a quantity of undeniable practical interest, namely how the expectation value of the spin measured along the $x$-, $y$-, and $z$-axes evolves in time, we can proceed on multiple routes, and the one I chose is not the fastest. However, it exploits the (possibly surprising) fact of the time evolution operator $U(t,0)$ obeying the same form of the Schrödinger equation as the state vectors (why?). Hence, we find its explicit form as

$$U(t,0) = \exp\left(\frac{-i\omega\sigma_z t}{\hbar}\right) \quad .$$

In the following, we will abbreviate $|z\pm\rangle \equiv |\pm\rangle$, and consider the expansion of the at $t_0 = 0$ prepared state

$$|A, t=0\rangle = c_+ |+\rangle + c_- |-\rangle \quad .$$

The to $t > 0$ evolved state is given by

$$U(t,0)\,|A,0\rangle = |A,t\rangle = c_+ \exp\left(\frac{-i\omega t}{2}\right)|+\rangle + c_- \exp\left(\frac{+i\omega t}{2}\right)|-\rangle \quad .$$

It is intuitive to identify a 3-dimensional vector with components given by the expectation values of the spin along the $x$-, $y$-, and $z$-axes, which tells us at any time what can be expected of measurements of an ensemble of identically prepared systems propagating through the magnetic field for a time $t$. For this quantity, we must compute the expectation values of the spin components along each axis, and we start with $x$:

$$\begin{aligned}
|\langle x\pm|A,t\rangle|^2 &= \left|\left[\left(\frac{1}{\sqrt{2}}\right)\langle +| \pm \left(\frac{1}{\sqrt{2}}\right)\langle -|\right]\cdot\left[\left(\frac{1}{\sqrt{2}}\right)e^{-i\omega t/2}|+\rangle + \left(\frac{1}{\sqrt{2}}\right)e^{+i\omega t/2}|-\rangle\right]\right|^2 \\
&= \left|\frac{1}{2}e^{-i\omega t/2} \pm \frac{1}{2}e^{+i\omega t/2}\right|^2 = \begin{cases} \cos^2\dfrac{\omega t}{2}\,, & \text{for } |x+\rangle \\ \sin^2\dfrac{\omega t}{2}\,, & \text{for } |x-\rangle \end{cases} \quad .
\end{aligned}$$

We have used the expansion of the eigenstates of $\sigma_x$ in terms of the eigenstates of $\sigma_z$ as deduced empirically in the first lecture as

$$|x\pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm |-\rangle) \quad .$$

With this result, we find

$$\langle\sigma_x\rangle = \left(\frac{\hbar}{2}\right)\cos^2\left(\frac{\omega t}{2}\right) + \left(\frac{-\hbar}{2}\right)\sin^2\left(\frac{\omega t}{2}\right) = \left(\frac{\hbar}{2}\right)\cos\omega t \quad ,$$

and similarly

$$\langle\sigma_y\rangle = \left(\frac{\hbar}{2}\right)\sin\omega t\,, \qquad \langle\sigma_z\rangle = 0 \quad .$$

I leave it up to you to square this with the commutation relations of the Pauli matrices with the Hamiltonian, and how to visualize this as a precession in real and Bloch-sphere space.

# $[P_{II}]$ Problem list

Do not get frightened by the following problems. They do not go beyond the concepts developed thus far, but their context is different and uses jargon from quantum computing. To solve the problems, the meaning and interpretation of the states and operators is not necessary, but you will gain more if you follow those links below which I inserted for further reading.

## Level 2

**P 3. A gate is a Hamiltonian switched on for a time $\tau$.** Qubit basis $|0\rangle \equiv |z+\rangle$, $|1\rangle \equiv |z-\rangle$; two qubits $|ab\rangle = |a\rangle \otimes |b\rangle$, the first one the control. The gates (do not confuse these $U$ with the time-evolution operator)

$$U_{\mathrm{NOT}} = \sigma_x, \qquad U_{\mathrm{Had}} = \tfrac{1}{\sqrt{2}}(\sigma_x + \sigma_z), \qquad U_{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes \sigma_x$$

are to be read as $U(\tau) = e^{-iH\tau/\hbar}$ for some Hermitian $H$, up to a global phase.

(a) For any Hermitian operator $A$ with $A^2 = \mathbf{1}$, show $e^{-i\alpha A} = \cos\alpha\, \mathbf{1} - i \sin\alpha\, A$.

(b) Let $H = \frac{\hbar\Omega}{2}\sigma_x$. Find the smallest $\tau$ with $U(\tau) = U_{\mathrm{NOT}}$ up to a phase, and give that phase. Which state (see Week I) is $U(\tau/2)\,|0\rangle$ equal to?

(c) Read the same $H$ as hopping between two bins $|0\rangle, |1\rangle$ in the form of Eq. (3-4): identify $w$ and the on-site terms $v_0, v_1$.

(d) For $H = \frac{\hbar\Omega}{2}\,\hat{n}\cdot\vec{\sigma}$, use $\frac{d}{dt}\langle L\rangle = \frac{i}{\hbar}\langle [H, L]\rangle$ and $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$ to show $\frac{d}{dt}\langle\vec{\sigma}\rangle = \Omega\,\hat{n}\times\langle\vec{\sigma}\rangle$. Compare this with what was discussed in class.

(e) Show $U_{\mathrm{Had}}^2 = \mathbf{1}$ and hence $U_{\mathrm{Had}} = i\, e^{-i\frac{\pi}{2}U_{\mathrm{Had}}}$. Give $H$, and describe the Hadamard gate as a precession on the Bloch sphere: axis and angle. Where do $|z\pm\rangle$ go?

(f) Show $U_{\mathrm{CNOT}} = \mathbf{1} - 2P$ with $P = |1\rangle\langle 1| \otimes |x-\rangle\langle x-|$ a projector, hence $U_{\mathrm{CNOT}} = e^{-i\pi P}$ and

$$H = \frac{\pi\hbar}{4\tau}(\mathbf{1}\otimes\mathbf{1} - \sigma_z\otimes\mathbf{1} - \mathbf{1}\otimes\sigma_x + \sigma_z\otimes\sigma_x).$$

Which of these terms cannot be generated by a field acting on each qubit separately?

(g) *Extension.* A processor applies gates one after another. Show $e^{-i(A+B)\Delta t} = e^{-iA\Delta t}e^{-iB\Delta t} + \frac{1}{2}[A, B]\Delta t^2 + O(\Delta t^3)$. Split the hopping matrix $W$ of the $N$-bin line into even and odd bonds, and turn $e^{-iWt}$ into a circuit of two-bin gates of type (b). How does the error scale with the number of steps?

[Nielsen & Chuang (10th anniv. ed.), Eqs. 4.4–4.8 and Ex. 4.2 for (a), (d); §4.7, Thm. 4.3 and Eq. 4.103 for (h).]

## Level 3

**P 5. The proton that draws the picture.** A proton in a field $\mathbf{B}$: $H = -\gamma_p\,\boldsymbol{\sigma}\cdot\mathbf{B}$, with $\gamma_p/2\pi = 42.58\ \mathrm{MHz\,T^{-1}}$, $\gamma_p > 0$.

(a) $\mathbf{B} = B_0\hat{z}$. Introduce the form $H = \omega\frac{\hbar}{2}\sigma_z$ by correctly identifying $\omega$, sign included. Does the proton precess in the same sense as the electron? You may now find the (Larmor) frequency $f_0 = \omega_0/2\pi$ at 1.5 and 3 T being of the same order as the FM radio band, 87.5–108 MHz.

(b) Now, one individual proton, measured along $\hat{x}$, yields $\pm\hbar/2$, but some detector which would detect the ensuing signal sees the summed moment of all protons. Take the populations of $|z\pm\rangle$ as Boltzmann, $\propto e^{-E_\pm/k_BT}$. Show $(N_+ - N_-)/N = \tanh(\hbar\omega_0/2k_BT)$ and evaluate it at 3 T, 310 K. One mm$^3$ of water holds $6.7\times10^{19}$ protons; how many are

net aligned? In the frame rotating with the protons take, as given, $H_{\text{rot}} = -\gamma_p B_1 \frac{\hbar}{2}\sigma_x$. Starting from $|z+\rangle$, find the time $t_{90}$ after which the state lies on the equator of the Bloch sphere, and which equator state it is. Evaluate for $B_1 = 5.9\ \mu\text{T}$.

(c) Switch $B_1$ off. Back in the laboratory, compute $\langle\sigma_x\rangle(t)$, $\langle\sigma_y\rangle(t)$ for the state of (c). This rotating moment induces the signal in the coil.

(d) Imaging: add a gradient, $B_z = B_0 + Gx$. In the language of the $[C_{II}]$ sheet, which $v_{\sigma\sigma'}(x)$ is this? Show that the signal of a proton density $\rho(x)$, with the carrier at $\omega_0$ removed, is $\int dx\, \rho(x)\, e^{\pm i\gamma_p G x t}$: a Fourier transform at $k = \gamma_p G t$. At $G = 10\ \text{mT}\,\text{m}^{-1}$, by how many hertz do two points 1 mm apart differ?

(e) *Extension.* Field inhomogeneity gives each proton its own offset $\delta$, and the summed moment of (d) decays. A $\pi$ pulse about $\hat{x}$ at time $T$ maps the azimuth $\phi \to -\phi$. Show that every proton is back in phase at $2T$, whatever its $\delta$. What does the echo not undo?

[Sakurai §2.1, "Spin Precession"; Nielsen & Chuang §7.7, refocusing in §7.7.3; E. L. Hahn, Phys. Rev. **80**, 580 (1950), for the echo.]

## $[T_{II}]$ The I-never-read-textbooks issues

- Let us milk as much out of the framework developed thus far by considering a one-dimensional line with $N$ sites as the state space. Let us now play good, and furnish the world a particle will experience as it dictates is time evolution.

  Your only task shall be to conjure an "intersting" Hamiltonian which is subjected only to the constraint of hermiticity.

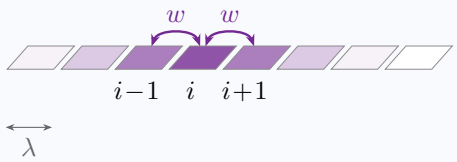
  Interesting questions in this context are to me:

  If the spin is a vector operator, can one understand the various bins as possible states of an $N$-dimensional spin?

[$C_{II}$] **Paulo slide**

## Quantum Mechanics I – Week II: Multi-dimensional extension and time evolution

### Position, discrete 1D band



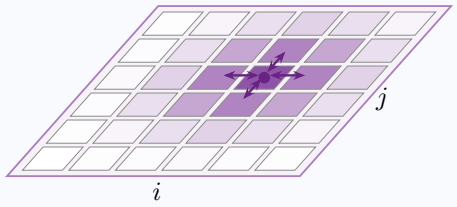

$$\psi_i = \langle i|\Psi\rangle$$

$$\sum_i |\psi_i|^2 = 1$$

$$i\,\partial_t \psi_i = \sum_{i'} W_{ii'}\,\psi_{i'}$$

$W_{i'i} = W^*_{ii'}$ 2 neighbours

### Position, discrete 2D lattice




$$\psi_{ij} = \langle i,j|\Psi\rangle\ , \quad |i,j\rangle = |i\rangle \otimes |j\rangle$$

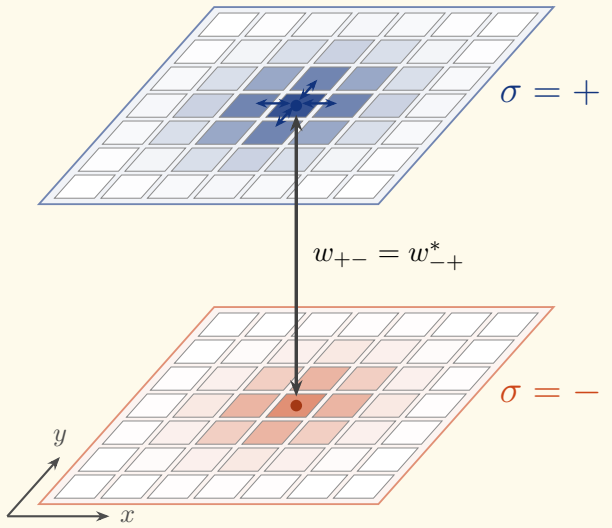
$$\sum_{i,j} |\psi_{ij}|^2 = 1$$

$$i\,\partial_t \psi_{ij} = \sum_{i'j'} W_{ij,i'j'}\,\psi_{i'j'}$$

4 neighbours

### 2-state system (qubit) ⊗ Position, discrete 2D lattice

$$|\Psi\rangle = \sum_{i,j,\sigma} \psi_{ij\sigma}\ |i,j\rangle \otimes |\sigma\rangle$$




$$\begin{pmatrix} \psi_{11+} \\ \psi_{12+} \\ \vdots \\ \psi_{ij+} \\ \vdots \\ \psi_{11-} \\ \psi_{12-} \\ \vdots \\ \psi_{ij-} \\ \vdots \end{pmatrix} \begin{matrix} \} \\ \\ \\ \\ \\ \} \\ \\ \\ \\ \\ \end{matrix} \begin{matrix} |+\rangle \\ \\ \\ \\ \\ |-\rangle \\ \\ \\ \\ \\ \end{matrix}$$

4 neighbours + 1 across

---

$\lambda \to 0$

$$\psi_i \to \sqrt{\lambda}\,\psi(x)$$

$$\int dx\,|\psi|^2 = 1$$

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\,\partial_x^2\psi + v\,\psi$$

$$\psi_{ij} \to \lambda\,\psi(x,y)$$

$$\int dx\,dy\,|\psi|^2 = 1$$

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\,\nabla^2\psi + v\,\psi$$

$$\psi_{ij\sigma} \to \lambda\,\psi_\sigma(x,y) = \langle x,y;\sigma|\Psi\rangle$$

$$\sum_\sigma \int dx\,dy\,|\psi_\sigma|^2 = 1$$

$$i\hbar\,\partial_t\psi_\sigma = -\frac{\hbar^2}{2m}\,\nabla^2\psi_\sigma + \sum_{\sigma'} v_{\sigma\sigma'}\,\psi_{\sigma'}$$

$$\Psi(x,y,t) = \begin{pmatrix} \psi_+(x,y,t) \\ \psi_-(x,y,t) \end{pmatrix}$$

$v = \omega S_z$: $e^{-i\omega t/2}$, $e^{+i\omega t/2}$

---

**recipe for a Schrödinger ket:**

$$H \xrightarrow[\text{guess, borrow, or steal an H}]{} |\Psi(0)\rangle \xrightarrow[\text{solve the eigenvalue problem}]{\text{prepare an initial state}} H\,|E_j\rangle = E_j\,|E_j\rangle \xrightarrow{\text{compute initial coefficients}} \alpha_j(0) = \langle E_j|\Psi(0)\rangle \xrightarrow[\text{evolve the state in time}]{} |\Psi(t)\rangle = \sum_j \alpha_j(0)\,e^{-\frac{i}{\hbar}E_j t}\,|E_j\rangle \xrightarrow{\text{compute measurement probabilities}} P_\lambda(t) = |\langle\lambda|\Psi(t)\rangle|^2$$

## $[\aleph_{III}]$ Uncertainty

### Complete sets of commuting observables

$$L\,|\lambda_i, \mu_a\rangle = \lambda_i\,|\lambda_i, \mu_a\rangle$$

$$M\,|\lambda_i, \mu_a\rangle = \mu_a\,|\lambda_i, \mu_a\rangle$$

$$L\,|\lambda, \mu\rangle = \lambda\,|\lambda, \mu\rangle\ , \qquad M\,|\lambda, \mu\rangle = \mu\,|\lambda, \mu\rangle$$

$$LM\,|\lambda, \mu\rangle = L\mu\,|\lambda, \mu\rangle = \lambda\mu\,|\lambda, \mu\rangle$$

$$LM\,|\lambda, \mu\rangle = ML\,|\lambda, \mu\rangle$$

$$[L, M]\,|\lambda, \mu\rangle = 0$$

$$[L, M] = 0$$

### Wave functions

$$|\Psi\rangle = \sum_{a,b,c,\ldots} \psi(a, b, c, \ldots)\,|a, b, c, \ldots\rangle$$

$$\psi(a, b, c, \ldots) = \langle a, b, c, \ldots|\Psi\rangle$$

$$P(a, b, c, \ldots) = \psi^*(a, b, c, \ldots)\,\psi(a, b, c, \ldots)$$

$$\psi(z+) = \langle z+|\Psi\rangle\ , \qquad \psi(z-) = \langle z-|\Psi\rangle$$

$$\psi(x+) = \langle x+|\Psi\rangle\ , \qquad \psi(x-) = \langle x-|\Psi\rangle$$

$$\sum_{a,b,c,\ldots} \psi^*(a, b, c, \ldots)\,\psi(a, b, c, \ldots) = 1$$

$$|\Psi\rangle = \sum_j \alpha_j\,|\psi_j\rangle$$

**Measurement**

$$L = \begin{pmatrix} r & w \\ w^* & r' \end{pmatrix}$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$L = a\,\sigma_x + b\,\sigma_y + c\,\sigma_z + d\,\mathbf{1}\,, \qquad a, b, c, d \in \mathbb{R}$$

$$[\sigma_x, \sigma_y] = 2i\sigma_z\,, \qquad [\sigma_y, \sigma_z] = 2i\sigma_x\,, \qquad [\sigma_z, \sigma_x] = 2i\sigma_y$$

**The uncertainty principle**

**The meaning of uncertainty**

$$\langle\Psi|A|\Psi\rangle = \sum_a a\,P(a)$$

$$\bar{A} = A - \langle A\rangle$$

$$\bar{A} = A - \langle A\rangle\,\mathbf{1}$$

$$\bar{a} = a - \langle A\rangle$$

$$(\Delta A)^2 = \sum_a \bar{a}^2\,P(a)$$

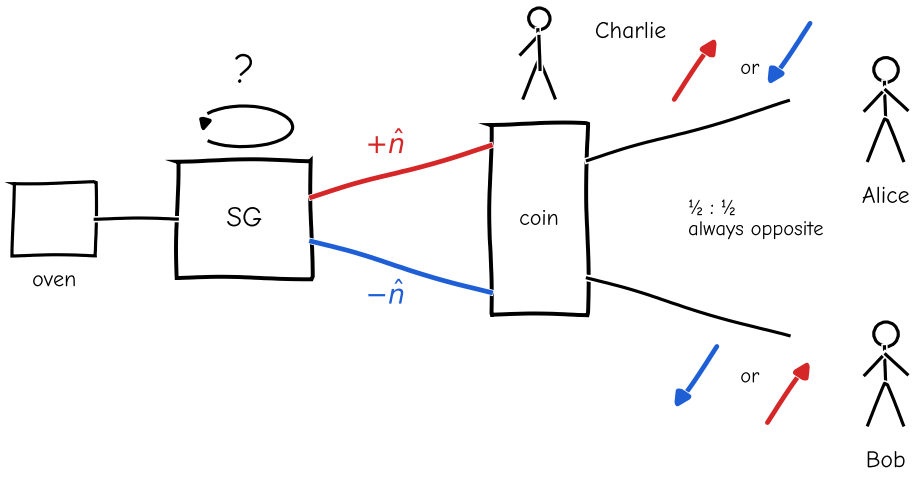
$$(\Delta A)^2 = \sum_a (a - \langle A\rangle)^2 P(a)$$

$$(\Delta A)^2 = \langle\Psi|\bar{A}^2|\Psi\rangle$$

$$(\Delta A)^2 = \langle\Psi|A^2|\Psi\rangle \qquad \text{for } \langle A\rangle = 0$$

## $[\beth_{III}]$ A 2-component system

Firstly, is there a difference between a classical Charlie preparing a pair of atoms in always opposite spin states and handing them randomly to Alice and Bob, and his quantum analogue? In other words, could he tell with certainty which spins states he hands to Alice and Bob without disturbing the system? The situation is sketched below.




$$(\text{Alice}, \text{Bob}) = (u,d) \text{ or } (d,u)\,, \qquad P(u,d) = P(d,u) = \tfrac{1}{2}$$

$$\text{Alice: } |\hat{n}+\rangle\,, \qquad \text{Bob: } |\hat{n}-\rangle\,, \qquad \hat{n} \text{ unknown, even to Charlie}$$

Charlie does not know the orientation of SG but is able to separate the system to hand one part to Alice and the other to Bob. Alice ($\sigma$) and Bob ($\tau$) make measurements along the same axis $\hat{n}$, and regardless of how far they are apart, the moment they measure their spin, e.g., Alice finds $\sigma(\hat{n}) = +1$, she knows with certainty that $\tau(\hat{n}) = -1$. Does this violate the law that no information can be transmitted faster than the speed of light?

What is the expectation value of the spin measurements for Alice and Bob individually? What is expected for their joint measurements?

$$\langle\sigma(\hat{a})\rangle = \tfrac{1}{2}\,\hat{a}\cdot\hat{n} + \tfrac{1}{2}(-\hat{a}\cdot\hat{n}) = 0\,, \qquad \Big\langle\tau(\hat{b})\Big\rangle = 0\,, \qquad P(\sigma(\hat{a}) = \pm 1) = \tfrac{1}{2}$$

$$\Big\langle\sigma(\hat{a})\,\tau(\hat{b})\Big\rangle = -(\hat{a}\cdot\hat{n})\Big(\hat{b}\cdot\hat{n}\Big)$$

$$P(\sigma(\hat{a}) = -\tau(\hat{a})) = \frac{1+(\hat{a}\cdot\hat{n})^2}{2} \qquad \Big(= 1 \text{ for } \hat{a} = \pm\hat{n}\,, \quad = \tfrac{1}{2} \text{ for } \hat{a}\perp\hat{n}\Big)$$

[$\hat{a}$, $\hat{b}$: Alice's and Bob's analyzer axes; $\hat{n}$: the SG axis, fixed; the coin decides who receives $|\hat{n}+\rangle$.]

[How does all this change if Charlie makes a measurement along an axis $\hat{m}$ of Alice's particle before handing it to her?]

Some formal considerations on how to employ our formalism for general composite systems:

What does it mean to have a 2-component quantum system? I must be able to analyze its parts individually but also to assess properties which result only as a consequence of their compositeness.

Note, the parts could, but do not have to be identical. As the generic example, consider a two-qubit system, and a qubit-tetrahedron system. List all possible states and use the following notation for the composite system.

$$\{uu,\ ud,\ du,\ dd\}\,, \qquad 2\times 2 = 4$$

$$\{u1,\ u2,\ u3,\ u4,\ d1,\ d2,\ d3,\ d4\}\,, \qquad 2\times 4 = 8$$

$$|ab\rangle\ , \qquad a \text{ Alice's label}, \quad b \text{ Bob's label}$$

[$1,\ldots,4$: the faces of the tetrahedron; its quantum counterpart is any 4-level system, e.g. spin $\frac{3}{2}$.]

Now promote the states to vectors, and accept the following equivalent notations:

$$|a\rangle \otimes |b\rangle = |a\rangle\,|b\rangle = |a,b\rangle = |ab\rangle$$

$$|uu\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |ud\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |du\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |dd\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

$$\begin{pmatrix}\alpha_u\\ \alpha_d\end{pmatrix} \otimes \begin{pmatrix}\beta_u\\ \beta_d\end{pmatrix} = \begin{pmatrix}\alpha_u\beta_u\\ \alpha_u\beta_d\\ \alpha_d\beta_u\\ \alpha_d\beta_d\end{pmatrix}$$

$$\big\langle a'b'\big|ab\big\rangle = \delta_{a'a}\,\delta_{b'b}$$

The missing postulate: The state space of a composite system is the tensor product of the component's state spaces.

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B\,, \qquad \dim \mathcal{H}_{AB} = N_A N_B$$

$$|\Psi\rangle = \sum_{a,b} \psi(a,b)\,|ab\rangle\ , \qquad \sum_{a,b} \psi^*(a,b)\,\psi(a,b) = 1$$

$$|\Psi\rangle = (\alpha_u\,|u\rangle + \alpha_d\,|d\rangle) \otimes (\beta_u\,|u\rangle + \beta_d\,|d\rangle)\,, \qquad \psi(a,b) = \alpha_a\beta_b$$

Can you show the following equivalence in both directions?

$$\psi(u,u)\,\psi(d,d) - \psi(u,d)\,\psi(d,u) = 0 \quad \Longleftrightarrow \quad |\Psi\rangle \text{ is a product state}$$

For two qubits, note that $\psi(a,b)$ can be represented as a $2\times 2$ matrix, which has rank one if the corresponding state is a product state.

How do we represent observables? Let us play a little with what expectation values look like for composite systems.

$$\sigma_i \equiv \sigma_i \otimes \mathbf{1}\,, \qquad \tau_i \equiv \mathbf{1} \otimes \tau_i\,, \qquad [\sigma_i, \tau_j] = 0$$

$$\sigma_z\,|ud\rangle = |ud\rangle\ , \quad \sigma_x\,|ud\rangle = |dd\rangle\ , \quad \sigma_y\,|ud\rangle = i\,|dd\rangle\ ,$$

$$\tau_z\,|ud\rangle = -\,|ud\rangle\ , \quad \tau_x\,|ud\rangle = |uu\rangle\ , \quad \tau_y\,|ud\rangle = -i\,|uu\rangle$$

$$\sigma_z \otimes \mathbf{1} = \mathrm{diag}(1,1,-1,-1)\,, \qquad \mathbf{1} \otimes \tau_z = \mathrm{diag}(1,-1,1,-1)$$

$$|\Psi\rangle = |\alpha\rangle \otimes |\beta\rangle : \qquad \langle\sigma_i\rangle = \langle\alpha|\sigma_i|\alpha\rangle\,, \qquad \langle\tau_j\rangle = \langle\beta|\tau_j|\beta\rangle$$

$$\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1\,, \qquad \langle\tau_x\rangle^2 + \langle\tau_y\rangle^2 + \langle\tau_z\rangle^2 = 1$$

$$\langle\sigma_i\tau_j\rangle = \langle\sigma_i\rangle\,\langle\tau_j\rangle$$

$$C_{ij} = \langle\sigma_i\tau_j\rangle - \langle\sigma_i\rangle\,\langle\tau_j\rangle$$

[$\langle X\rangle \equiv \langle\Psi|X|\Psi\rangle$; $C_{ij}$: correlation of Alice's $\sigma_i$ with Bob's $\tau_j$, zero in every product state.]

The fundamental question about a composite system: Does the specification of the state vector imply knowledge of the measurement outcomes of the individual spins? In other words, can I infer the microscopic structure (the individual component states) from a bulk measurement, which means something which depends on the how both parts of the system are realized together? In statistical language, we would phrase the difference as the one between marginal and joint distributions: the probability to have two random variables to assume simultaneously certain values vs. the probability of one to assume a value regardless of the other.

Let us consider this on the example of the so-called singlet state[2]:

$$|\mathrm{sing}\rangle = \tfrac{1}{\sqrt{2}}(|ud\rangle - |du\rangle) \quad ,$$

yields the following expectation values for the individual spins:

$$\langle\sigma_i\rangle = \langle\tau_i\rangle = 0\,, \qquad i = x,y,z \quad ,$$

while looking at the expected outcome if we measure them jointly and multiply the results:

$$\langle\sigma_i\tau_j\rangle = -\delta_{ij}\,, \qquad C_{ij} = -\delta_{ij} \quad \text{… fascinating!}$$

The question is now how the individual measurements and the correlation one are affected–if at all–by rotating the measurement axes of both spins into arbitrary and thus possibly-different directions compared to that axis which was used to define the original singlet state. It needs some effort, but calculating representative expectation values for rotated axes is quite rewarding.

Now, let us spend more lines on the measurement of $|\mathrm{sing}\rangle$. Specifically, on the operator whose eigenstate $|\mathrm{sing}\rangle$ is. To this end, consider

$$\vec{\sigma}\cdot\vec{\tau} = \sigma_x\tau_x + \sigma_y\tau_y + \sigma_z\tau_z \quad ,$$

and convince yourself that in addition to $|\mathrm{sing}\rangle$, the triplet of states

$$|T_1\rangle = \tfrac{1}{\sqrt{2}}(|ud\rangle + |du\rangle)\,, \quad |T_2\rangle = \tfrac{1}{\sqrt{2}}(|uu\rangle + |dd\rangle)\,, \quad |T_3\rangle = \tfrac{1}{\sqrt{2}}(|uu\rangle - |dd\rangle) \quad ,$$

are its eigenstates:

[2] In the following, be careful to identify in which state averages are taken.

$$\vec{\sigma}\cdot\vec{\tau}\,|\text{sing}\rangle = -3\,|\text{sing}\rangle\ , \qquad \vec{\sigma}\cdot\vec{\tau}\,|T_k\rangle = +\,|T_k\rangle \quad . \tag{1}$$

At this point, it is worth remembering the relation between the operators and possible measurement outcomes. In particular, how to think of a sum, $\vec{\sigma}_z\otimes\mathbf{1}+\mathbf{1}\otimes\vec{\tau}_z$, vs. the product, $\vec{\sigma}_z\otimes\vec{\tau}_z$, of spin measurements on the two particles. Although we have not explicitly mentioned time, the correct interpretation of the former is a measurement that analyzes the properties of one component of the system, only, before/after/simultaneously with another measurement on teh other component but *always* on identical copies/states of the system. The product, however, looks how certain combinations of outcomes are present in the state of the system. A simple example are the different outcomes:

$$\sigma_z\tau_z\,|\text{sing}\rangle = -\,|\text{sing}\rangle\ , \qquad \text{vs.} \qquad (\sigma_z\otimes\mathbf{1}+\mathbf{1}\otimes\vec{\tau}_z)\,|\text{sing}\rangle = 0 \quad .$$

Another sanity check for our understanding of how to employ QM postulates to multi-component systems is to express the unit operator in various bases:

$$\mathbf{1} = |\text{sing}\rangle\langle\text{sing}| + \sum_{k=1}^{3}|T_k\rangle\langle T_k| \quad , \tag{2}$$

$$\mathbf{1} = \sum_{\alpha,\beta\in\{u,d\}}|\alpha\beta\rangle\langle\alpha\beta| \quad , \tag{3}$$

with, e.g., $P_{\text{sing}} = |\text{sing}\rangle\langle\text{sing}| = \frac{1}{4}(\mathbf{1}-\vec{\sigma}\cdot\vec{\tau})$, and where the factors of $|\alpha\beta\rangle$ do not necessarily have to be eigenstates of identical $\sigma$ orientation.

Regarding the classification/labeling of states with possible measurement outcomes of a complete list of commuting operators, it is seemingly surprising that

$$[\sigma_i\tau_i,\sigma_j\tau_j] = 0\,, \quad \text{while} \quad [\sigma_i,\sigma_j] = 2i\epsilon_{ijk}\sigma_k \quad \text{for} \quad i,j,k\in\{x,y,z\} \quad .$$

So for the individual part, one 2-valued label – which can be the outcome of a spin measurement along any axis – suffices, but although the three operators $\sigma_i\tau_i$ commute, they fail to distinguish between the four entangled states $|\text{sing}\rangle, |T_1\rangle, |T_2\rangle, |T_3\rangle$.

[Does this failure of a set of commuting operators to distinguish between the elements of one orthogonal basis imply that it fails in this sense for all orthogonal bases?]

After having discussed fomal aspects of a system combined from two two-level systems, it seems natural to ask how specific states of it can be prepared with the machinery we employed for an individual system, i.e., a polarization filter/Stern-Gerlach apparatus.

$$(A\otimes B)(|\alpha\rangle\otimes|\beta\rangle) = A\,|\alpha\rangle\otimes B\,|\beta\rangle$$

$$P_{\text{sing}}\,|ud\rangle = \tfrac{1}{\sqrt{2}}\,|\text{sing}\rangle\ , \qquad P = \tfrac{1}{2}$$

$$H = J\,\vec{\sigma}\cdot\vec{\tau}: \qquad e^{-iHt/\hbar}\,|ud\rangle\Big|_{Jt/\hbar=\pi/8} = \tfrac{1}{2}\,e^{-i\pi/8}[(1+i)\,|ud\rangle + (1-i)\,|du\rangle]$$

[every Stern–Gerlach filter and every local evolution has the form $A\otimes B$: product in, product out. A filter on the non-product $\vec{\sigma}\cdot\vec{\tau}$, or an interaction coupling the two spins, is needed; the last state has

$|\psi(u,u)\psi(d,d) - \psi(u,d)\psi(d,u)| = \frac{1}{2}$, maximal. In the laboratory: sources with total $J = 0$, e.g. an atomic cascade.]

Discuss the two essential types of operators along the two types of states: product and entangled states, and, independent and correlated operators.

$$\text{local: } A \otimes \mathbf{1}\,,\ \mathbf{1} \otimes B\,; \qquad \text{product: } A \otimes B\,; \qquad \text{non-product: } \sum_k A_k \otimes B_k \neq A \otimes B$$

| | $\lvert\alpha\rangle \otimes \lvert\beta\rangle$ | $\lvert\text{sing}\rangle$ |
|---|---|---|
| local $\sigma_i$ | $\langle\alpha\lvert\sigma_i\rvert\alpha\rangle$ | 0 |
| product $\sigma_i\tau_j$ | $\langle\sigma_i\rangle\,\langle\tau_j\rangle$ | $-\delta_{ij}$ |
| non-product $\vec{\sigma}\cdot\vec{\tau}$ | $\geq -1$ | $-3$ |

## Project: maximal entanglement beyond two parts

[Three qubits: $\sigma_i^{(k)}$ acts on qubit $k$ only.]

$$\text{whole: } |\Psi\rangle \text{ known completely;} \qquad \text{parts: } \langle\vec{\sigma}\rangle = \langle\vec{\tau}\rangle = 0$$

1. Is this a definition of *entangled* or of *maximally* entangled? Show, for every two-qubit state,
$$|\langle\vec{\sigma}\rangle|^2 = |\langle\vec{\tau}\rangle|^2 = 1 - 4|\psi(u,u)\psi(d,d) - \psi(u,d)\psi(d,u)|^2$$
and say what lies between the two ends of the scale.
2. "Everything known about the whole": which commuting observables have $|\text{sing}\rangle$ as their only common eigenvector? Are any of them local?
3. Lift the definition to three qubits and test it on
$$|\text{GHZ}\rangle = \tfrac{1}{\sqrt{2}}(|uuu\rangle + |ddd\rangle)\,, \qquad |W\rangle = \tfrac{1}{\sqrt{3}}(|uud\rangle + |udu\rangle + |duu\rangle)\,.$$
Compute $\langle\vec{\sigma}^{(k)}\rangle$. If $|W\rangle$ fails, is it less entangled, or entangled differently?
4. A pair is a part, too. Compute $\langle\sigma_z^{(1)}\sigma_z^{(2)}\rangle$ and $\langle\sigma_x^{(1)}\sigma_x^{(2)}\rangle$ in both states. Measure qubit 3 along $z$, then instead along $x$: what is left of qubits 1 and 2? Whom does the GHZ pair resemble, the singlet or the classical Charlie?
5. Can a three-qubit state leave *every* part, single qubits and pairs, maximally unknown? If not, what does the lifted definition have to give up? Four qubits?
6. Charlie hands out three atoms, each with a card carrying predetermined readings $m_x^{(k)}, m_y^{(k)} = \pm 1$. Check
$$\sigma_x^{(1)}\sigma_x^{(2)}\sigma_x^{(3)}\,|\text{GHZ}\rangle = +\,|\text{GHZ}\rangle\ , \qquad \sigma_x^{(1)}\sigma_y^{(2)}\sigma_y^{(3)}\,|\text{GHZ}\rangle = -\,|\text{GHZ}\rangle\ ,$$
and the two cyclic partners of the second, and that all four operators commute. Multiply the cards. Does the contradiction need statistics, an inequality, or one run? Compare with the three boxes of Chapter I.

## BB84

[Photon version: polarizers at 0°/90° and 45°/135°.]

[SIM] `bb84.html`

$$\text{bit } 0: \ |z+\rangle\,,\ |x+\rangle\ ; \qquad \text{bit } 1: \ |z-\rangle\,,\ |x-\rangle$$

1. Alice: random bit, random basis $z$ or $x$, sends the state.
2. Bob: random basis, measures.
3. Public channel: bases only; keep equal bases, $P = \frac{1}{2}$ — the sifted key.
4. Publish a random subset of it, estimate the error rate; abort above threshold, otherwise error correction and privacy amplification.

$$|\langle x\pm|z\pm\rangle|^2 = \tfrac{1}{2}\,, \qquad [\sigma_z, \sigma_x] \neq 0$$

$$\text{Eve, intercept–resend:} \qquad P_{\text{err}} = \underbrace{\tfrac{1}{2}}_{\text{wrong basis}} \cdot \underbrace{\tfrac{1}{2}}_{\text{Bob random}} = \tfrac{1}{4}$$

[threshold $\approx 11\%$ for one-way post-processing. Eve cannot copy the unknown state instead: no-cloning.]

## $[R_{III}]$ Reading list


[7] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing", Theoretical Computer Science **560**, 7–11 (2014) doi:10.1016/j.tcs.2014.05.025, reprint of Proc. IEEE Int. Conf. on Computers, Systems and Signal Processing, Bangalore (1984), p. 175.

[8] C. H. Bennett, F. Bessette, G. Brassard, L. Salvail, and J. Smolin, "Experimental quantum cryptography", Journal of Cryptology **5**, 3–28 (1992) doi:10.1007/BF00191318.

[9] A. Aspect, P. Grangier, and G. Roger, "Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell's Inequalities", Physical Review Letters **49**, 91–94 (1982) doi:10.1103/PhysRevLett.49.91.

[10] J.-W. Pan, D. Bouwmeester, M. Daniell, H. Weinfurter, and A. Zeilinger, "Experimental test of quantum nonlocality in three-photon Greenberger-Horne-Zeilinger entanglement", Nature **403**, 515–519 (2000) doi:10.1038/35000514.

[11] L. Susskind and A. Friedman, *Quantum Mechanics: The Theoretical Minimum* (Basic Books, 2014), lectures 6–7 [pedagogical].

[12] N. D. Mermin, "Quantum mysteries revisited", American Journal of Physics **58**, 731–734 (1990) doi:10.1119/1.16503, the GHZ argument without inequalities [pedagogical].

[13] L. E. Ballentine, *Quantum Mechanics: A Modern Development*, 2nd ed. (World Scientific, 2014), doi:10.1142/9038, chapter 20, "Bell's Theorem and Its Consequences" [pedagogical].

## $[P_{III}]$ Problem list

**Level 1**

**Level 2**

**Level 3**

## $[T_{III}]$ The I-never-read-textbooks issues

$[C_{III}]$ **Paulo slide**

# $[\aleph_{IV}]$ Working with continuous functions

## discrete

[eigenvalues $\lambda_i$ of $L$; grid $x_n = n\varepsilon$, $n \in \mathbb{Z}$, $\varepsilon$: bin width]

## continuous

[$x \in \mathbb{R}$; $\varepsilon \to 0$]

## Wave function

$$|\Psi\rangle = \sum_i \psi(\lambda_i)\,|\lambda_i\rangle$$

$$|\Psi\rangle = \int dx\,\psi(x)\,|x\rangle$$

$$\psi(\lambda_i) = \langle\lambda_i|\Psi\rangle$$

$$\psi(x) = \langle x|\Psi\rangle$$

$$|\Psi\rangle \;\leftrightarrow\; \begin{pmatrix}\psi(\lambda_1)\\ \psi(\lambda_2)\\ \vdots\end{pmatrix}$$

$$|\Psi\rangle \;\leftrightarrow\; \psi:\ \mathbb{R}\to\mathbb{C}\,,\quad x\mapsto\psi(x)$$

## Functions are vectors

$$(z\,u + w\,v)_n = z\,u_n + w\,v_n\,,\qquad z,w\in\mathbb{C}$$

$$(z\,\psi + w\,\phi)(x) = z\,\psi(x) + w\,\phi(x)$$

$$\langle\Psi| \;\leftrightarrow\; (\psi^*(\lambda_1),\ \psi^*(\lambda_2),\ \ldots)$$

$$\langle\Psi| \;\leftrightarrow\; \psi^*(x)$$

## Integrals replace sums

$$\sum_n \varepsilon\,f(x_n)$$

$$\longrightarrow \int dx\,f(x)$$

$$\psi_n \equiv \langle n|\Psi\rangle = \sqrt{\varepsilon}\,\psi(x_n)\,,\qquad |x_n\rangle = \frac{1}{\sqrt{\varepsilon}}\,|n\rangle$$

$$\psi(x) = \langle x|\Psi\rangle$$

[$|n\rangle$: particle in bin $n$, $\langle n|m\rangle = \delta_{nm}$]

[same $\psi(x)$ as above; the $\sqrt{\varepsilon}$ stays with the discrete side]

$$\langle\Psi|\Phi\rangle = \sum_n \psi_n^*\phi_n = \sum_n \varepsilon\,\psi^*(x_n)\,\phi(x_n)$$

$$\langle\Psi|\Phi\rangle = \int_{-\infty}^{\infty} dx\,\psi^*(x)\,\phi(x)$$

## Densities replace probabilities

$$P(\lambda_i) = \psi^*(\lambda_i)\,\psi(\lambda_i)$$

$$P(x) = \psi^*(x)\,\psi(x)$$

[per unit length]

$$P(n) = |\psi_n|^2 = \varepsilon\,|\psi(x_n)|^2$$

$$\frac{P(n)}{\varepsilon} \;\longrightarrow\; P(x)$$

$$\sum_{a\le x_n\le b} P(n)$$

$$P(a,b) = \int_a^b dx\,P(x) = \int_a^b dx\,\psi^*(x)\,\psi(x)$$

$$\sum_i \psi^*(\lambda_i)\,\psi(\lambda_i) = 1$$

$$\int_{-\infty}^{\infty} dx\,\psi^*(x)\,\psi(x) = 1$$

## Dirac delta replaces Kronecker delta

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\delta(x - x') = 0 \quad (x \neq x')\,, \qquad \int dx\,\delta(x) = 1$$

$$\sum_j \delta_{ij}\,F_j = F_i$$

$$\int_{-\infty}^{\infty} dx'\,\delta(x - x')\,F(x') = F(x)$$

$$\langle \lambda_i | \lambda_j \rangle = \delta_{ij}$$

$$\langle x | x' \rangle = \delta(x - x')$$

$$\langle x_n | x_m \rangle = \frac{\delta_{nm}}{\varepsilon}$$

$$\longrightarrow\ \delta(x_n - x_m)$$

$$\sum_n |n\rangle\,\langle n| = \sum_n \varepsilon\ |x_n\rangle\,\langle x_n| = \mathbf{1}$$

$$\int dx\ |x\rangle\,\langle x| = \mathbf{1}$$

$\frac{\delta_{nm}}{\varepsilon}$ : height $\frac{1}{\varepsilon}$ on one bin, area 1

$$\delta(x) = \lim_{\nu\to\infty} \frac{\nu}{\sqrt{\pi}}\,e^{-(\nu x)^2}$$

[width $1/\nu$, height $\nu/\sqrt{\pi}$, area 1]

## Cartesian basis

$$(e_n)_m = \delta_{nm}$$

$$e_n\ \longrightarrow\ \delta_x\,, \qquad \delta_x(x') = \delta(x - x')$$

$$f_n = \langle f, e_n \rangle = \sum_m f_m (e_n)_m$$

$$f(x) = \langle f, \delta_x \rangle = \int dx'\,f(x')\,\delta_x(x')$$

$$\langle f, g \rangle = \sum_n f_n\,g_n$$

$$\langle f, g \rangle = \int dx\,f(x)\,g(x)$$

[real $f$, $g$]

## Function $\to$ functional

$$F(f)\,, \qquad F:\ \mathbb{R}^N \to \mathbb{R}$$

$$F[f]$$

[a number for every function $f$]

$$\partial_{f_n} F(f) = \lim_{\eta\to 0} \frac{1}{\eta}[F(f + \eta\,e_n) - F(f)]$$

$$\frac{\delta F[f]}{\delta f(x)} = \lim_{\eta\to 0} \frac{1}{\eta}(F[f + \eta\,\delta_x] - F[f])$$

$$dF_f(g) = \sum_n \partial_{f_n} F(f)\ g_n$$

$$DF_f[g] = \int dx\,\frac{\delta F[f]}{\delta f(x)}\,g(x)$$

[grid: $\delta F/\delta f(x_n) = \varepsilon^{-1}\,\partial_{f_n} F$]

$$\frac{\partial f_m}{\partial f_n} = \delta_{nm}$$

$$\frac{\delta f(y)}{\delta f(x)} = \delta(x - y)$$

$$F = \sum_n \varepsilon\, f_n^2 \;: \qquad \frac{1}{\varepsilon}\,\partial_{f_n} F = 2 f_n$$

$$F[f] = \int dx\, f(x)^2 \;: \qquad \frac{\delta F}{\delta f(x)} = 2 f(x)$$

## Integration by parts

$$\frac{G_{n+1} - G_n}{\varepsilon}$$

$$\longrightarrow \quad \frac{dG}{dx}$$

$$\sum_n F_n (G_{n+1} - G_n) = -\sum_n (F_n - F_{n-1})\, G_n$$

$$d(FG) = F\, dG + G\, dF$$

$$\int_a^b F\, dG = FG\Big|_a^b - \int_a^b G\, dF$$

$$\int_{-\infty}^{\infty} dx\, F\, \frac{dG}{dx} = -\int_{-\infty}^{\infty} dx\, \frac{dF}{dx}\, G$$

[$F_n, G_n \to 0$ for $|n| \to \infty$; relabel $n+1 \to n$ in one sum]

[$F, G \to 0$ for $|x| \to \infty$]

$$\sum_n \varepsilon\, F_n\, \frac{G_{n+1} - G_n}{\varepsilon} = -\sum_n \varepsilon\, \frac{F_n - F_{n-1}}{\varepsilon}\, G_n$$

[forward difference on $G$ becomes backward difference on $F$]

## Linear operators

$$L[z\,\psi(x) + w\,\phi(x)] = z\, L\psi(x) + w\, L\phi(x)$$

$$X\psi(x) = x\,\psi(x)$$

$$D\psi(x) = \frac{d\psi(x)}{dx}$$

$$L^\dagger = L \quad \Longleftrightarrow \quad \langle\Psi|L|\Phi\rangle = \langle\Phi|L|\Psi\rangle^*$$

$$\langle\Psi|X|\Phi\rangle = \int dx\, \psi^*(x)\, x\, \phi(x)\,, \qquad \langle\Phi|X|\Psi\rangle = \int dx\, \phi^*(x)\, x\, \psi(x)$$

$$x \in \mathbb{R} \quad \Longrightarrow \quad X^\dagger = X$$

$$\langle\Psi|D|\Phi\rangle = \int dx\, \psi^*(x)\, \frac{d\phi(x)}{dx}\,, \qquad \langle\Phi|D|\Psi\rangle = \int dx\, \phi^*(x)\, \frac{d\psi(x)}{dx}$$

$$\langle\Phi|D|\Psi\rangle = -\int dx\, \psi(x)\, \frac{d\phi^*(x)}{dx}$$

[integration by parts]

$$\langle\Psi|D|\Phi\rangle = -\langle\Phi|D|\Psi\rangle^* \qquad \Longrightarrow \qquad D^\dagger = -D$$

[anti-Hermitian]

$$(-i\hbar D)^\dagger = -i\hbar D\,, \qquad -i\hbar D\,\psi(x) = -i\hbar\,\frac{d\psi(x)}{dx}$$

## The state of a particle

$$\text{classical: } (x,p)\,, \qquad p = m\dot{x}$$

$$|x,p\rangle\,, \quad \psi(x,p) = \langle x,p|\Psi\rangle \qquad \text{no such basis}$$

[like $\sigma_z$ and $\sigma_x$: $x$ or $p$, not both]

### position

### Operator

$$X\psi(x) = x\,\psi(x)$$

### Eigenvalues and eigenvectors

$$X\,|\Psi\rangle = x_0\,|\Psi\rangle$$

$$(x - x_0)\,\psi(x) = 0$$

$$\psi_{x_0}(x) = \delta(x - x_0)\,, \qquad x_0 \in \mathbb{R}$$

$$\langle x|x'\rangle = \delta(x - x')$$

### Waves

$$\delta(x - x_0)$$

[particle at $x_0$]

### Wave function

$$\psi(x) = \langle x|\Psi\rangle$$

### momentum

$$P = -i\hbar D\,, \qquad P\psi(x) = -i\hbar\,\frac{d\psi(x)}{dx}$$

[$\hbar$: $D$ has units $1/L$, $P$ has $ML/T$]

$$P\,|\Psi\rangle = p\,|\Psi\rangle$$

$$\frac{d\psi(x)}{dx} = \frac{ip}{\hbar}\,\psi(x)$$

$$\psi_p(x) = A\,e^{ipx/\hbar}\,, \qquad A = \frac{1}{\sqrt{2\pi\hbar}}$$

$$\langle p|p'\rangle = |A|^2 \int dx\, e^{i(p'-p)x/\hbar} = |A|^2\, 2\pi\hbar\,\delta(p - p')$$

[$\langle p|p'\rangle = \delta(p - p')$ fixes $A$; the source's $A = 1/\sqrt{2\pi}$ holds for $\hbar = 1$ only]

$$e^{ip(x+2\pi\hbar/p)/\hbar} = e^{ipx/\hbar}\,e^{2\pi i} = e^{ipx/\hbar}$$

$$\lambda = \frac{2\pi\hbar}{p}$$

[$\lambda$: wavelength]

$$\tilde{\psi}(p) = \langle p|\Psi\rangle$$

$$P(x) = \psi^*(x)\,\psi(x)$$

$$\mathbf{1} = \int dx\ |x\rangle\,\langle x|$$

$$P(p) = |\langle p|\Psi\rangle|^2 = \tilde{\psi}^*(p)\,\tilde{\psi}(p)$$

$$\mathbf{1} = \int dp\ |p\rangle\,\langle p|$$

## Change of representation

$$\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}}\,e^{-ipx/\hbar}$$

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ipx/\hbar}$$

$[|\langle x|p\rangle|^2$ independent of $x$ and $p$, as $|\langle z\pm|x\pm\rangle|^2 = \frac{1}{2}]$

$$\tilde{\psi}(p) = \int dx\ \langle p|x\rangle\,\langle x|\Psi\rangle$$

$$\psi(x) = \int dp\ \langle x|p\rangle\,\langle p|\Psi\rangle$$

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}}\int dx\,e^{-ipx/\hbar}\,\psi(x)$$

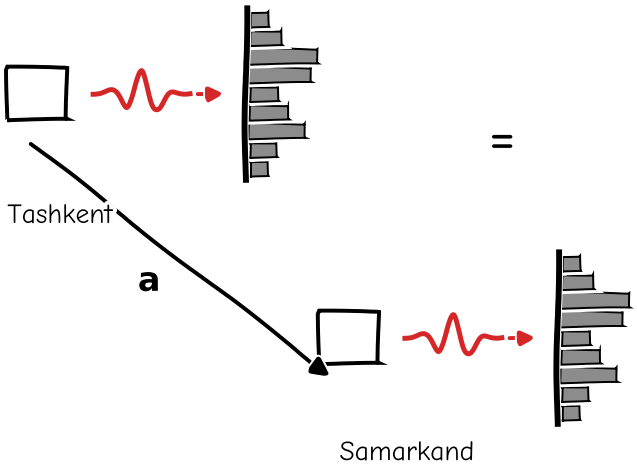
$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}}\int dp\,e^{ipx/\hbar}\,\tilde{\psi}(p)$$

## PRINCIPLE OF RELATIVITY

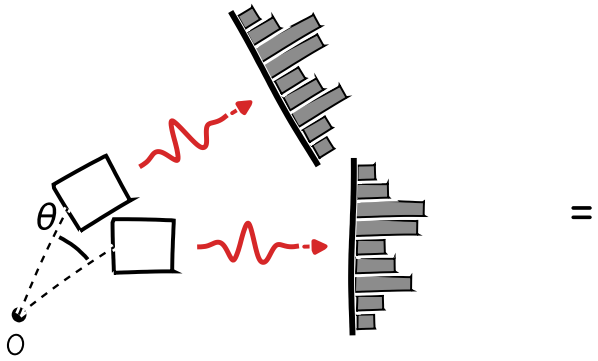
observers whose frames differ by $\mathbf{x} \to R\,\mathbf{x} + \mathbf{a} + \mathbf{v}\,t\,,\quad t \to t + s$ see identical physics: $|\langle \phi_n | \psi \rangle|^2 = |\langle \phi'_n | \psi' \rangle|^2$

$\Longrightarrow$ (Wigner) $|\psi'\rangle = U|\psi\rangle\,,\quad U$ unitary

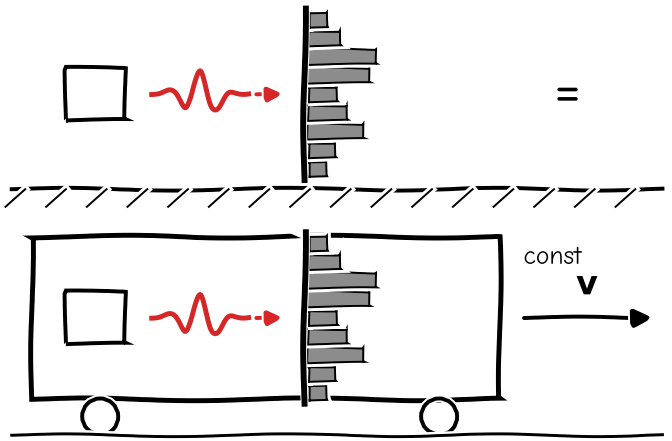
(a) $\mathbf{x} \to \mathbf{x} + \mathbf{a}$



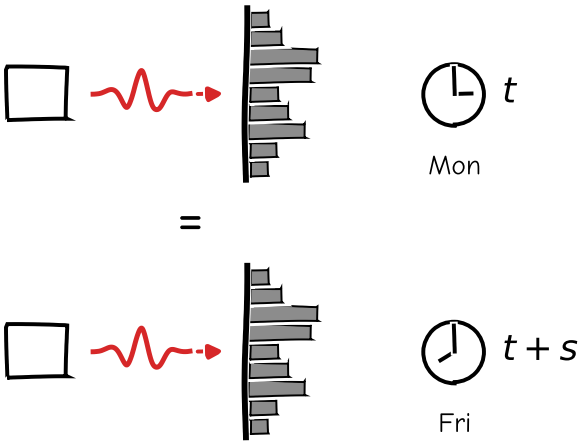

(b) $\mathbf{x} \to R\,\mathbf{x}$



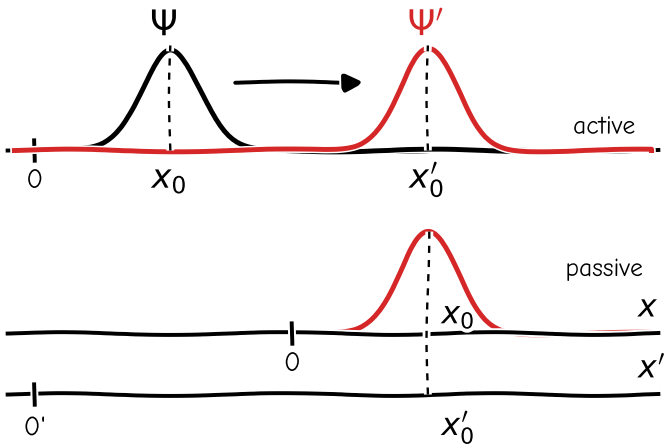

(c) $\mathbf{x} \to \mathbf{x} + \mathbf{v}\,t$



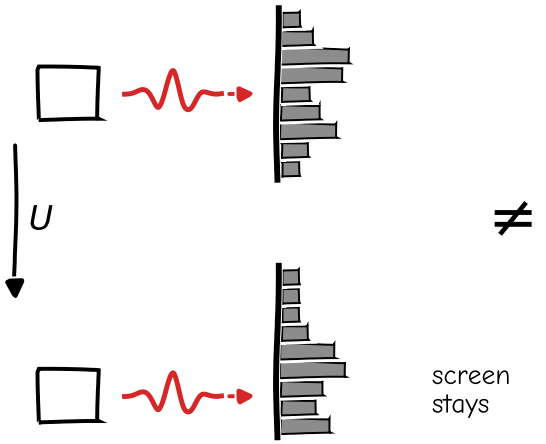

(d) $t \to t + s$




(e) $\Psi'(x') = \Psi(x)$




(f) $|\psi\rangle \to U|\psi\rangle$ only



# $[P_{IV}]$ Problem list

A Galilei transformation $\tau = (R, \mathbf{a}, \mathbf{v}, s)$ acts on an event $(\mathbf{x}, t)$ as $\mathbf{x} \to R\,\mathbf{x} + \mathbf{a} + \mathbf{v}\,t$, $t \to t + s$. Unitary operators as in the lecture: $e^{-i\theta J_\alpha}$ (rotation about $x_\alpha$), $e^{-iaP_\alpha}$ (displacement along $x_\alpha$), $e^{ivG_\alpha}$ (velocity along $x_\alpha$), $e^{isH}$ (time displacement). In a product of operators the rightmost acts first.

## Level 1

**P 1. Two frames in a row.**

(a) Apply $\tau_1$, then $\tau_2$. Show that the result is again a Galilei transformation $\tau_3 = \tau_2\tau_1$ and that

$$R_3 = R_2R_1, \qquad \mathbf{a}_3 = \mathbf{a}_2 + R_2\mathbf{a}_1 + \mathbf{v}_2 s_1, \qquad \mathbf{v}_3 = \mathbf{v}_2 + R_2\mathbf{v}_1, \qquad s_3 = s_2 + s_1.$$

(b) $\tau_1$: rotation by 90° about $x_3$, $\mathbf{a}_1 = (1,0,0)\,\mathrm{m}$, $\mathbf{v}_1 = (0,1,0)\,\mathrm{m\,s^{-1}}$, $s_1 = 2\,\mathrm{s}$. $\tau_2$: rotation by 90° about $x_1$, $\mathbf{a}_2 = (0,0,1)\,\mathrm{m}$, $\mathbf{v}_2 = (1,0,0)\,\mathrm{m\,s^{-1}}$, $s_2 = 1\,\mathrm{s}$. Find $R_3, \mathbf{a}_3, \mathbf{v}_3, s_3$. Check your answer by following the event $(\mathbf{x}, t) = (\mathbf{0}, 0)$ through $\tau_1$ and $\tau_2$ one at a time.

(c) Compute $\tau_1\tau_2$. Which of the four parameters differ from (b)? Apart from $R_2R_1 \neq R_1R_2$, which term in (a) is responsible?

(d) Find the inverse $\tau^{-1}$ of a general $\tau = (R, \mathbf{a}, \mathbf{v}, s)$, and check $\tau^{-1}\tau = (\mathbf{1}, \mathbf{0}, \mathbf{0}, 0)$ with (a).

(e) Pure boost $\mathbf{v}$ followed by pure time displacement $s$, and the reverse order: show that the two results differ by the displacement $\mathbf{v}s$. Compare with $[G_\alpha, H] = iP_\alpha$.

(f) *Extension.* $R_3$ in (b) is a single rotation. Find its axis and angle.

## Level 2

**P 2. The Jacobi identity fixes what geometry leaves open.** The method of the four-fold product gives commutators of generators only up to a multiple of the identity, e.g. $[J_\alpha, P_\beta] = i\epsilon_{\alpha\beta\gamma}P_\gamma + (?)\mathbf{1}$, $[P_\alpha, P_\beta] = (?)\mathbf{1}$, $[J_\alpha, P_\alpha] = (?)\mathbf{1}$, and the same with $G$ in place of $P$.

(a) For any three operators, show $[[A,B],C] + [[B,C],A] + [[C,A],B] = 0$, and bring it to the form

$$[[A,B],C] = [[C,B],A] + [[A,C],B].$$

(b) Show that a multiple of the identity in $[A,B]$ drops out of $[[A,B],C]$.

(c) Take $A = J_1$, $B = P_2$, $C = P_1$. Show $[P_3, P_1] = 0$: the unknown multiple is fixed.

(d) Repeat (c) with $G$ in place of $P$: $[G_3, G_1] = 0$.

(e) Take $A = J_1$, $B = G_2$ and $C = P_1$, then $C = P_3$, with $[G_\alpha, P_\beta] = (?)\mathbf{1}$. Show $[G_3, P_1] = 0$ and $[G_3, P_3] = [G_2, P_2]$, hence $[G_\alpha, P_\beta] = i\delta_{\alpha\beta}M\mathbf{1}$ with one real number $M$.

(f) Can $M$ be removed by $G_\alpha \to G_\alpha + b_\alpha\mathbf{1}$, $P_\alpha \to P_\alpha + c_\alpha\mathbf{1}$? Contrast with $[J_\alpha, J_\beta] = i\epsilon_{\alpha\beta\gamma}J_\gamma + i\epsilon_{\alpha\beta\gamma}b_\gamma\mathbf{1}$, where $J_\alpha + b_\alpha\mathbf{1} \to J_\alpha$ does remove it.

(g) *Extension.* Ballentine's version of (e) for $[G_\alpha, H] = iP_\alpha$ uses $A = J_1$, $B = G_2$, $C = H$. Carry it out, and say which multiple of the identity has to be known already.

**P 3. The Galilei group as $3 \times 3$ matrices.** One dimension, $R = 1$. Write an event as the column $(x, t, 1)^T$.

(a) Show that $x \to x + a + vt$, $t \to t + s$ is the matrix

$$T(a, v, s) = \begin{pmatrix} 1 & v & a \\ 0 & 1 & s \\ 0 & 0 & 1 \end{pmatrix},$$

and that $T(a_2, v_2, s_2)\, T(a_1, v_1, s_1)$ reproduces P 1(a).

(b) Define $A = \partial_a T$, $V = \partial_v T$, $S = \partial_s T$ at $a = v = s = 0$. Show $T(a, v, s) = e^{aA}\, e^{vV}\, e^{sS}$. Compute $e^{aA+vV+sS}$: which entry differs from $T(a, v, s)$?

(c) Compute $[V, S]$, $[V, A]$, $[A, S]$.

(d) Read the matrices as $P \leftrightarrow iA$, $G \leftrightarrow -iV$, $H \leftrightarrow -iS$, so that $e^{-iaP} \leftrightarrow e^{aA}$, $e^{ivG} \leftrightarrow e^{vV}$, $e^{isH} \leftrightarrow e^{sS}$. Compare (c) with $[G, H] = iP$, $[P, H] = 0$, $[G, P] = iM\mathbf{1}$. Which value of $M$ do the matrices give?

(e) *Extension.* Add a second space direction and rotations about $x_3$: $4 \times 4$ matrices on $(x_1, x_2, t, 1)^T$. Find the generator $\Omega$ of rotations, read $J_3 \leftrightarrow i\Omega$, and check $[J_3, P_\alpha] = i\epsilon_{3\alpha\gamma} P_\gamma$ and $[J_3, G_\alpha] = i\epsilon_{3\alpha\gamma} G_\gamma$.

### Level 3

**P 4. Commutators from four small steps.**

(a) Expand each factor to second order in $\eta$ and show

$$e^{i\eta A}\, e^{i\eta B}\, e^{-i\eta A}\, e^{-i\eta B} = \mathbf{1} + \eta^2 [B, A] + O(\eta^3).$$

Why does no $\eta^2 A^2$ or $\eta^2 B^2$ survive?

(b) $A = P_2$, $B = J_1$. Read the product right to left as four coordinate maps (rotation by $\eta$ about $x_1$, displacement $\eta$ along $x_2$, and their inverses). Show that to $O(\eta^2)$ the net map is $x_3 \to x_3 - \eta^2$. Which single unitary operator is this? Conclude $[J_1, P_2] = iP_3 + (?)\mathbf{1}$.

(c) $A = G_1$, $B = J_3$. Show that the net map is a velocity $\eta^2$ along $x_2$, and conclude $[J_3, G_1] = iG_2 + (?)\mathbf{1}$.

(d) $A = P_1$, $B = G_1$. Show that the four maps compose to the identity exactly. What does this tell you about $[G_1, P_1]$, and what does it not?

(e) *Extension.* With $[G_1, P_1] = iM\mathbf{1}$ from P 2(e), the product in (d) is $\mathbf{1} + i\eta^2 M$: the same frame, a different phase. Why is no experiment of the kind in panels (a)–(d) of the sketch able to detect this phase on a single state, and what kind of experiment could?

## [$\aleph_V$] Wave functions in position and momentum space

[Sakurai §1.7]

### Wave function

$$|\alpha\rangle = \int dx' \; |x'\rangle \langle x'|\alpha\rangle \;, \qquad \psi_\alpha(x') \equiv \langle x'|\alpha\rangle$$

$$\langle\beta|\alpha\rangle = \int dx' \, \psi^*_\beta(x') \, \psi_\alpha(x')$$

$$\langle\beta|A|\alpha\rangle = \int dx' \int dx'' \, \psi^*_\beta(x') \; \langle x'|A|x''\rangle \; \psi_\alpha(x'')$$

$$\langle\beta|f(x)|\alpha\rangle = \int dx'\,\psi^*_\beta(x')\,f(x')\,\psi_\alpha(x')$$

**Momentum operator in the position basis**

$$\mathcal{T}(\Delta x') = 1 - \frac{ip\,\Delta x'}{\hbar}\,, \qquad \mathcal{T}(\Delta x')\,\big|x'\big\rangle = \big|x' + \Delta x'\big\rangle$$

[translation, Sakurai §1.6]

$$\left(1 - \frac{ip\,\Delta x'}{\hbar}\right)|\alpha\rangle = \int dx'\ \big|x' + \Delta x'\big\rangle\,\langle x'|\alpha\rangle = \int dx'\ \big|x'\big\rangle\,\big\langle x' - \Delta x'\big|\alpha\big\rangle$$

$$\big\langle x' - \Delta x'\big|\alpha\big\rangle = \langle x'|\alpha\rangle - \Delta x'\,\frac{\partial}{\partial x'}\,\langle x'|\alpha\rangle$$

$$\langle x'|p|\alpha\rangle = -i\hbar\,\frac{\partial}{\partial x'}\,\langle x'|\alpha\rangle$$

$$\langle x'|p|x''\rangle = -i\hbar\,\frac{\partial}{\partial x'}\,\delta(x' - x'')$$

$$\langle\beta|p|\alpha\rangle = \int dx'\,\psi^*_\beta(x')\left(-i\hbar\,\frac{\partial}{\partial x'}\right)\psi_\alpha(x')$$

$$\big\langle x'\big|p^n\big|\alpha\big\rangle = (-i\hbar)^n\,\frac{\partial^n}{\partial x'^n}\,\langle x'|\alpha\rangle$$

**Momentum-space wave function**

$$p\,\big|p'\big\rangle = p'\,\big|p'\big\rangle\ , \qquad \big\langle p'\big|p''\big\rangle = \delta(p' - p'')$$

$$|\alpha\rangle = \int dp'\ \big|p'\big\rangle\,\langle p'|\alpha\rangle\ , \qquad \phi_\alpha(p') \equiv \langle p'|\alpha\rangle\ , \qquad \int dp'\,\big|\phi_\alpha(p')\big|^2 = 1$$

**Transformation function**

$$\langle x'|p|p'\rangle = -i\hbar\,\frac{\partial}{\partial x'}\,\langle x'|p'\rangle = p'\,\langle x'|p'\rangle \qquad \Rightarrow \qquad \langle x'|p'\rangle = N\,e^{ip'x'/\hbar}$$

$$\delta(x' - x'') = \int dp'\ \langle x'|p'\rangle\,\langle p'|x''\rangle = 2\pi\hbar\,|N|^2\,\delta(x' - x'') \qquad \Rightarrow \qquad \langle x'|p'\rangle = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ip'x'/\hbar}$$

$$\psi_\alpha(x') = \frac{1}{\sqrt{2\pi\hbar}}\int dp'\,e^{ip'x'/\hbar}\,\phi_\alpha(p')\,, \qquad \phi_\alpha(p') = \frac{1}{\sqrt{2\pi\hbar}}\int dx'\,e^{-ip'x'/\hbar}\,\psi_\alpha(x')$$

[cf. [$\aleph_{IV}$], change of representation]

In the reminder of this lecture, we talked about the Schrödinger equation as it applies to the abstract state vector and to the expansion components of it. The latter are referred to as the state's wave function. Depending on the choice of basis, the operators of the Hamiltonian assume different forms,and we used the spicific position basis to derive a probability-current density. Explicitely, we found the one-dimensional expression

$$\begin{aligned} j(x',t) &= \frac{\hbar}{2mi}\left[\psi^*(x',t)\frac{\partial}{\partial x'}\psi(x',t) - \psi(x',t)\frac{\partial}{\partial x'}\psi^*(x',t)\right] \\ &= \frac{1}{2m}\left[\psi^*(x',t)\hat{p}_x\psi(x',t) - \psi(x',t)\hat{p}_x\psi^*(x',t)\right] \end{aligned}$$

with the natural 3-dimensional generalization

$$\mathbf{j}(\mathbf{r},t) = \frac{\hbar}{2mi}\left[\psi^*(\mathbf{r},t)\nabla\psi(\mathbf{r},t) - \psi(\mathbf{r},t)\nabla\psi^*(\mathbf{r},t)\right] = \frac{1}{2m}\left[\psi^*(\mathbf{r},t)\hat{\mathbf{p}}\psi(\mathbf{r},t) - \psi(\mathbf{r},t)\hat{\mathbf{p}}\psi^*(\mathbf{r},t)\right] \quad .$$

## $[P_V]$ Problem list

### Level 1

**P 1. A free point mass.**

(a) Write down and solve the 1-dimensional Schrödinger equation for a free particle without spin and with mass $m$ for its general state and the corresponding wave function in the position representation.

(b) Solve this equation without imposing any boundary conditions.

(c) For each of the two linearly independent solutions, compute the corresponding probability current density $j(x',t)$.

(d) From the sign of the probability current density $j(x',t)$, interpret the direction of motion of the particle.

## $[\aleph_{VI}]$ 1-dimensional rectangular potentials

[Merzbacher §6]

### The step potential

### The well

### The barrier

## $[P_{VI}]$ Problem list

### Level X

In this project, you shall investigate bounds on the number of states that a reasonably arbitrarily shaped potential well can sutain. The problem is motivated by the dscussion given in A. Martin's article on "Bound states in the strong coupling limit" (Helv. Phys. Acta 45 (1972) 140-8).

The goal of this open-ended project is to develop an algorithm which allows you to increase the number of bound states of a certain potential well in a very intuitive fashion by adding/augmenting a well which supports $N$ such states by extending it via the addition of another rectangular well and thereby yielding a potential with precisely $N + 1$ bound states.

The procedure should be similar to building a structure with Lego blocks, and using this structure as a form to carve out a potential well with the desired number of bound states. The structure must certainly not contain any cavities or gaps.

**P 1.** Read the article and particularize each formula and argument to the 1-dimensional case.

**P 2.** Choose an exemplary potential which satisfies the Hö lder continuity condition. Give